

R Exercises for the OODA Short Course

IBC, Seoul, Korea

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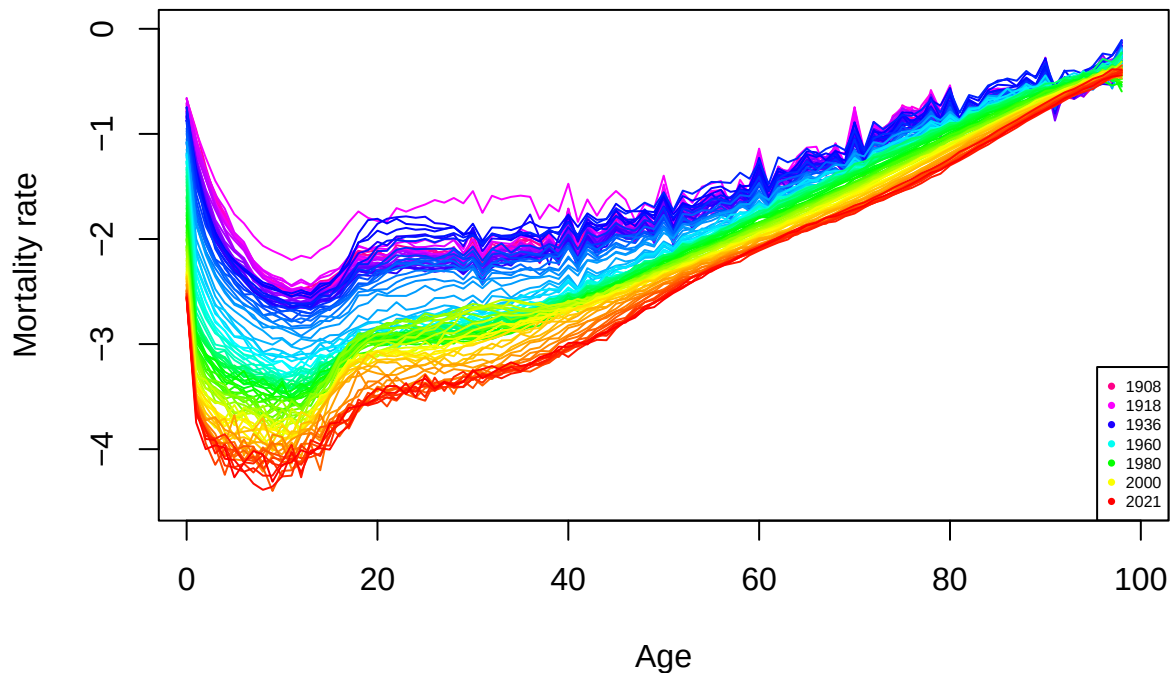
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R Exercise 1: Spanish Male Mortality data

Load the data for this exercise. Check that any external files are in your working directory before evaluating the chunk.

Slide context: R Exercise 1: Spanish Male Mortality data

```
x <- read.csv("Spain-Mx_1x1.csv")
x$Age[x$Age=="110+"]<-"110"
Age <- as.double(x$Age) ; Year <- as.double(x$Year)
Male <- as.double(x$Male) ; Female <- as.double(x$Female)
n<-114 ; d<-99
X <- matrix(0,n,d)
for (i in 1908:2021){
  for (j in 0:98){
    X[i-1907,j+1] <- Male[ (Age==j)&(Year==i)]
  }
}
Y <- log10(X)
palette(rainbow(n+10))
plot( Y[1,], type="n",xlab="Age",ylab="Mortality rate",ylim=c(-4.5,0))
for ( i in 1:n){
  lines(x$Age[1:d],Y[i,],col=n+1-i)
}
legend("bottomright",c("1908","1918","1936","1960","1980","2000","2021"),
      col=c(n,n-10,n-28,n-52,n-72,n-92,n-113),pch=16,cex=0.5)
```

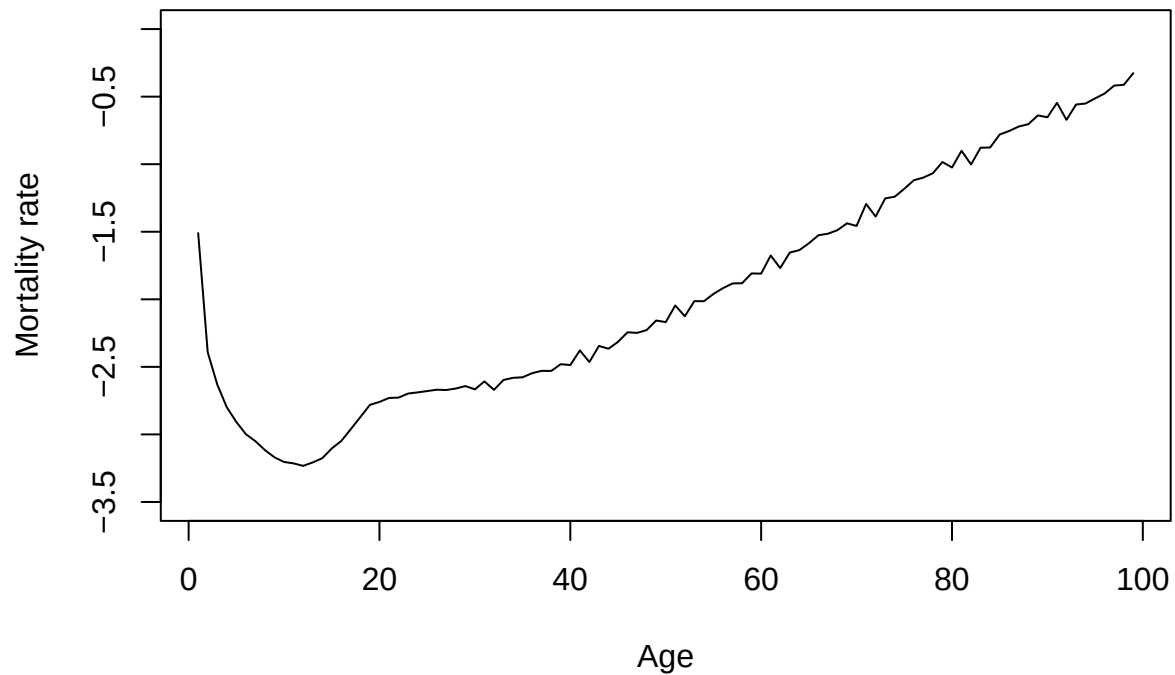


R Exercise 1: Residuals

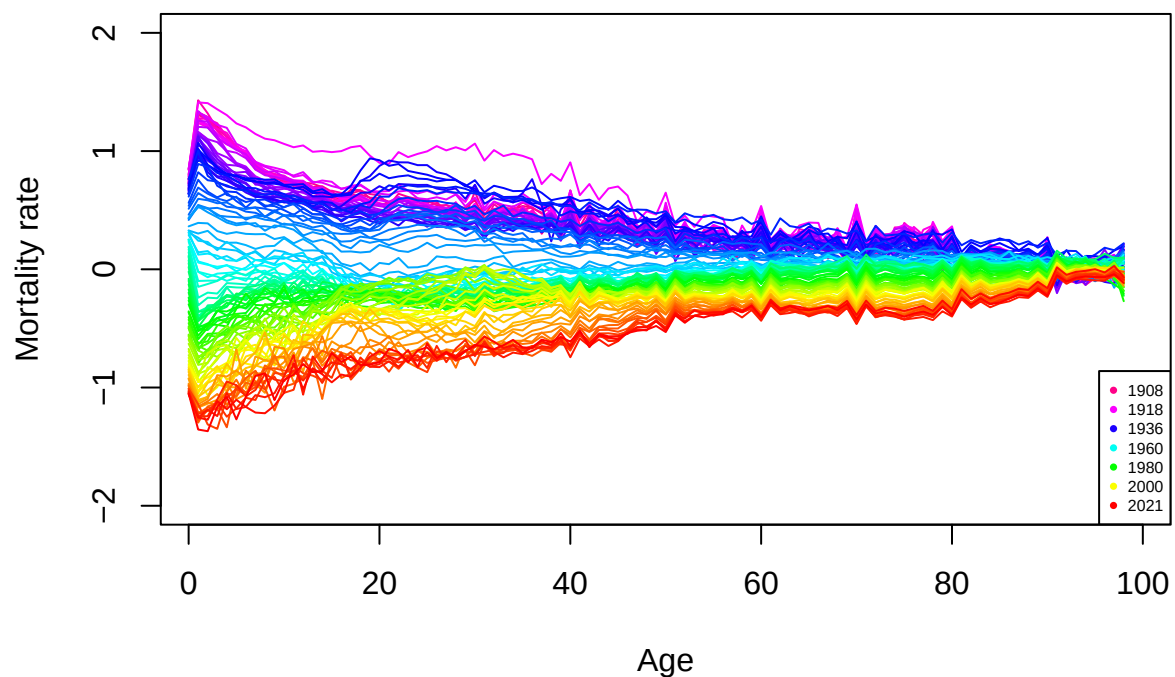
Create the first visual checks of the data or fitted objects. Use these plots to understand the structure before moving on to summaries.

Slide context: R Exercise 1: Residuals

```
Ybar <- apply( Y , 2 , mean)
plot( Ybar, type="l",xlab="Age",ylab="Mortality rate",ylim=c(-3.5,0))
```



```
palette(rainbow(n+10))
plot( Y[1,], type="n",xlab="Age",ylab="Mortality rate",ylim=c(-2,2))
for ( i in 1:n){
  lines(x$Age[1:d],Y[i,]-Ybar,col=n+1-i)
}
legend("bottomright",c("1908","1918","1936","1960","1980","2000","2021"),
      col=c(n,n-10,n-28,n-52,n-72,n-92,n-113),pch=16,cex=0.5)
```

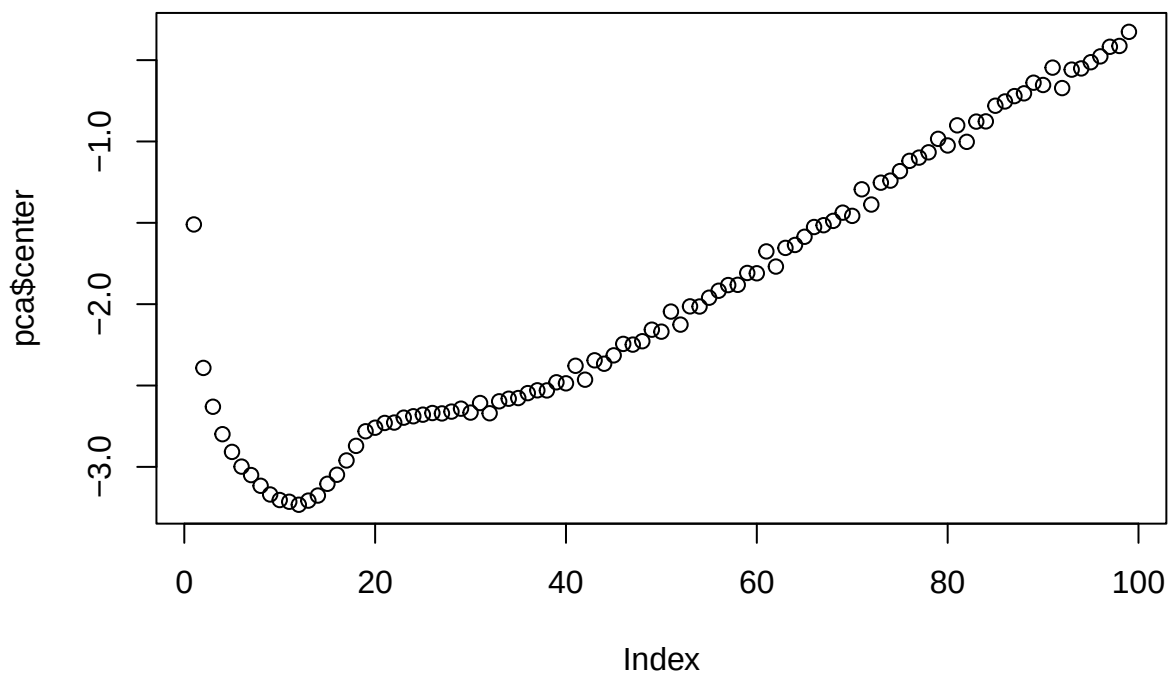


R Exercise 1: Mortality PCA

Create the first visual checks of the data or fitted objects. Use these plots to understand the structure before moving on to summaries.

Slide context: R Exercise 1: Mortality PCA

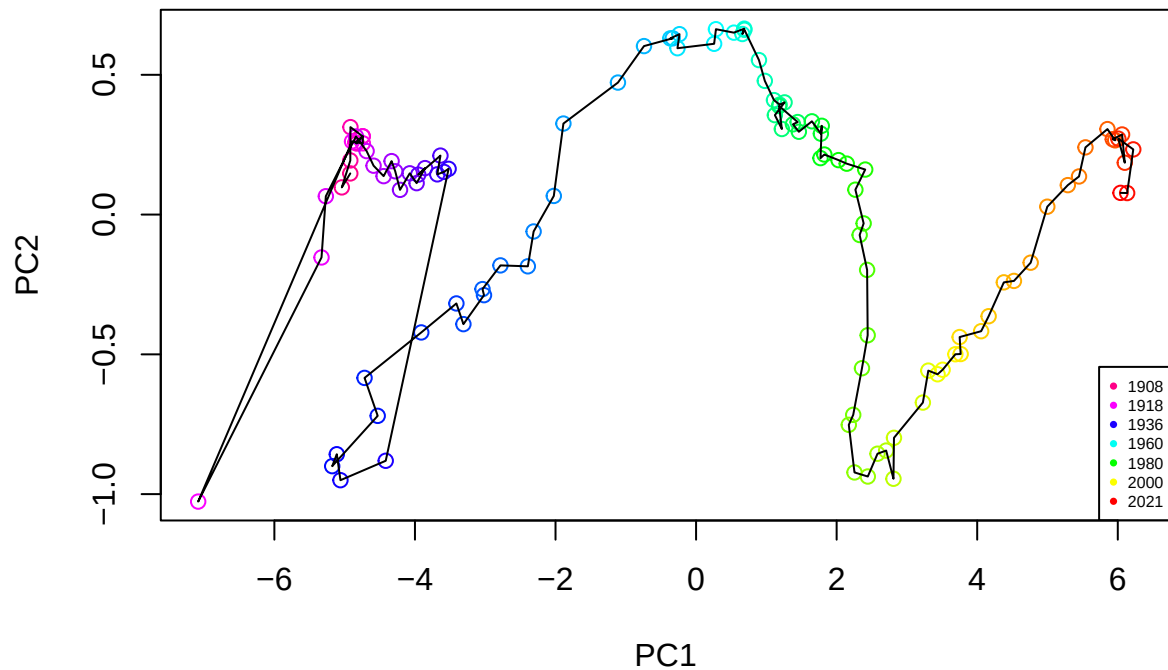
```
pca <- prcomp( Y )  
plot(pca$center)
```



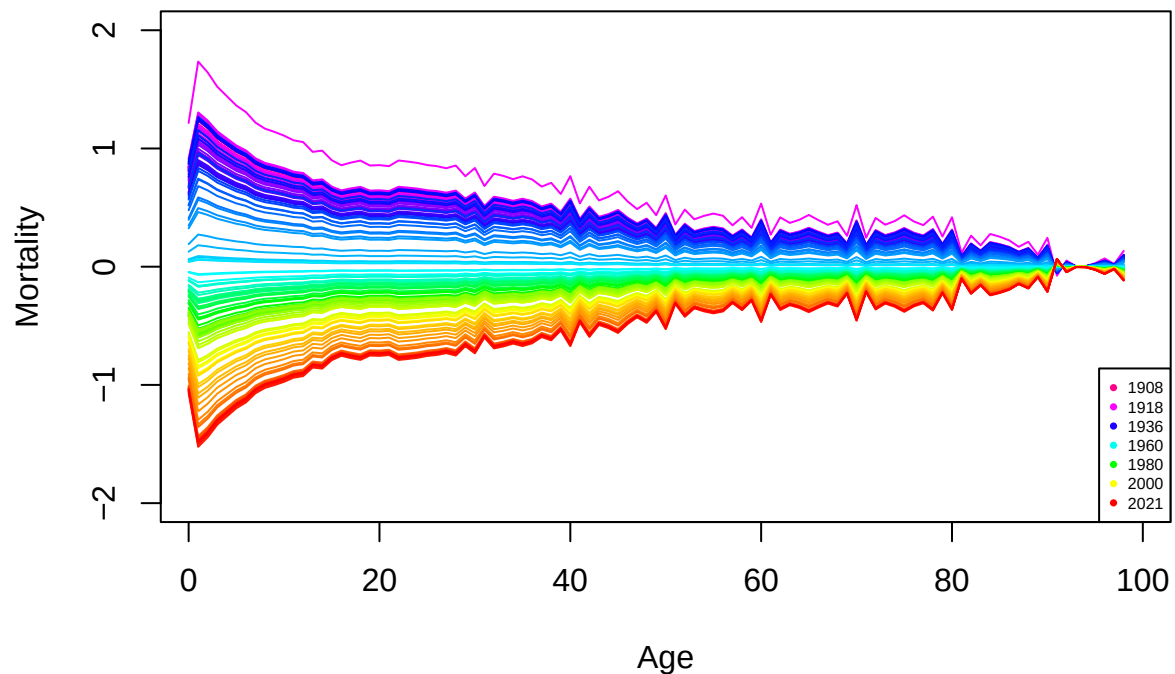
```
print(pca$sdev[1:2]**2/sum(pca$sdev**2)*100)
```

```
## [1] 96.831039  1.424947
```

```
plot(pca$x[,1],pca$x[,2],xlab="PC1",ylab="PC2",col=n:1)  
lines(pca$x[,1],pca$x[,2])  
legend("bottomright",c("1908","1918","1936","1960","1980","2000","2021"),  
      col=c(n,n-10,n-28,n-52,n-72,n-92,n-113),pch=16,cex=0.5)
```

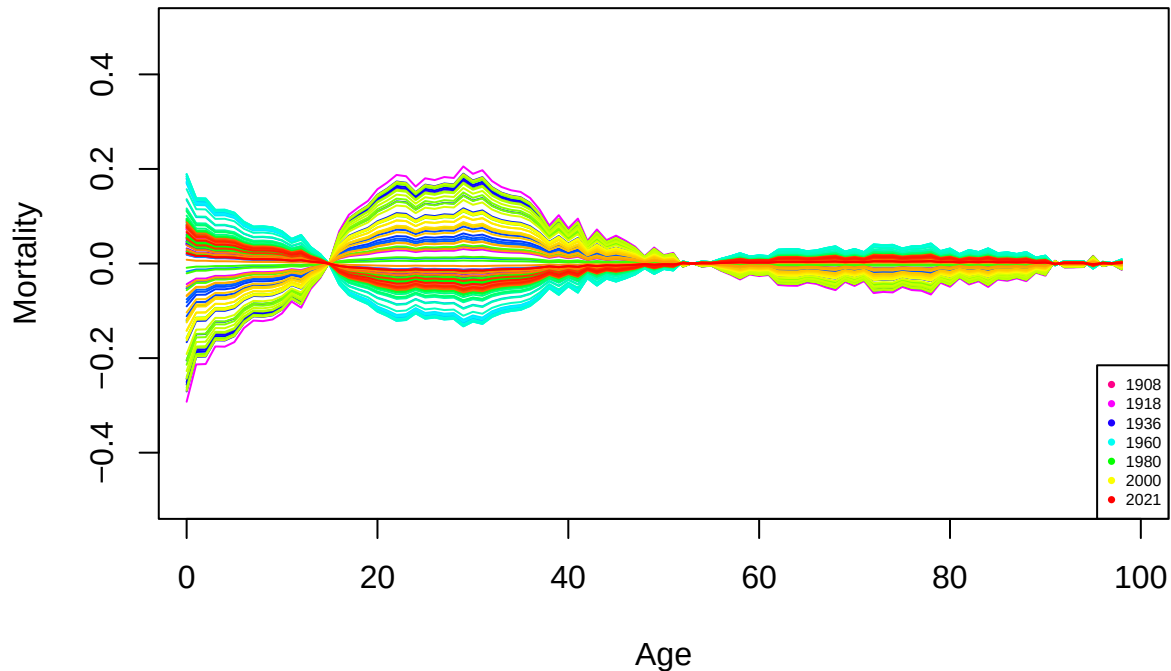


```
plot( Y[1,], type="n",xlab="Age",ylab="Mortality",ylim=c(-2,2))
for ( i in 1:n){
  lines(x$Age[1:d],pca$x[i,1]*pca$rotation[,1],col=n+1-i)
}
legend("bottomright",c("1908","1918","1936","1960","1980","2000","2021"),
      col=c(n,n-10,n-28,n-52,n-72,n-92,n-113),pch=16,cex=0.5)
```



```
plot( Y[1,], type="n",xlab="Age",ylab="Mortality",ylim=c(-0.5,0.5))
for ( i in 1:n){
  lines(x$Age[1:d],pca$x[i,2]*pca$rotation[,2],col=n+1-i)
}
```

```
legend("bottomright",c("1908","1918","1936","1960","1980","2000","2021"),
      col=c(n,n-10,n-28,n-52,n-72,n-92,n-113),pch=16,cex=0.5)
```



R Exercise 2: Pan Cancer data in R

Load the data for this exercise. Check that any external files are in your working directory before evaluating the chunk.

Slide context: The cancer groups are Bladder (BLCA), Kidney Renal (KIRC), Ovarian (OV), Head and Neck Squamous Cell (HNSC), Colon Adenocarcinoma (COAD), and Breast (BRCA). Pan Cancer data

```
palette("default")
x<-read.csv("PanCancer-ReducedGeneExpressionData.csv",header=TRUE)
dim(x)
```

```
## [1] 12479 301
```

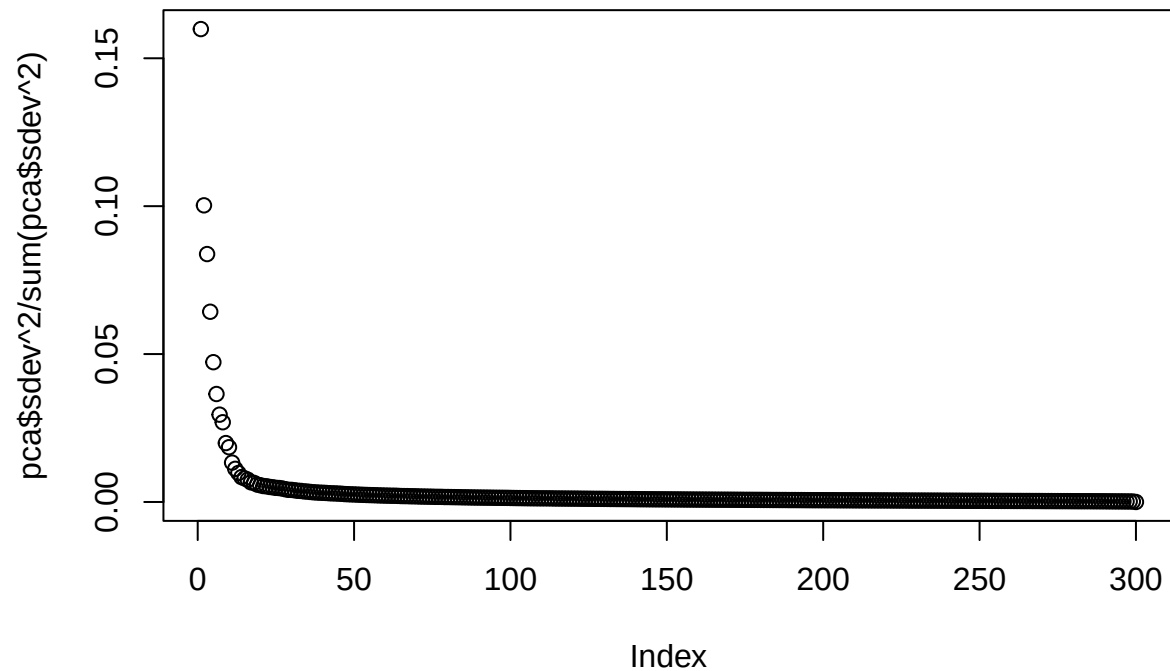
```
w<-x[2:12479,2:301] # remove first row and column
w<-sapply(w,as.numeric)
group<-unname(x[1,2:301])
group<-c(as.matrix(group))
table(group)
```

```
## group
## BLCA BRCA COAD HNSC KIRC OV
## 50 50 50 50 50 50
```

```
gcol<-rep(1:6,each=50) # give each group a different color
X<-t(w)
dim(X)
```

```
## [1] 300 12478
```

```
pca <- prcomp(X)
plot(pca$sdev**2/sum(pca$sdev**2))
```

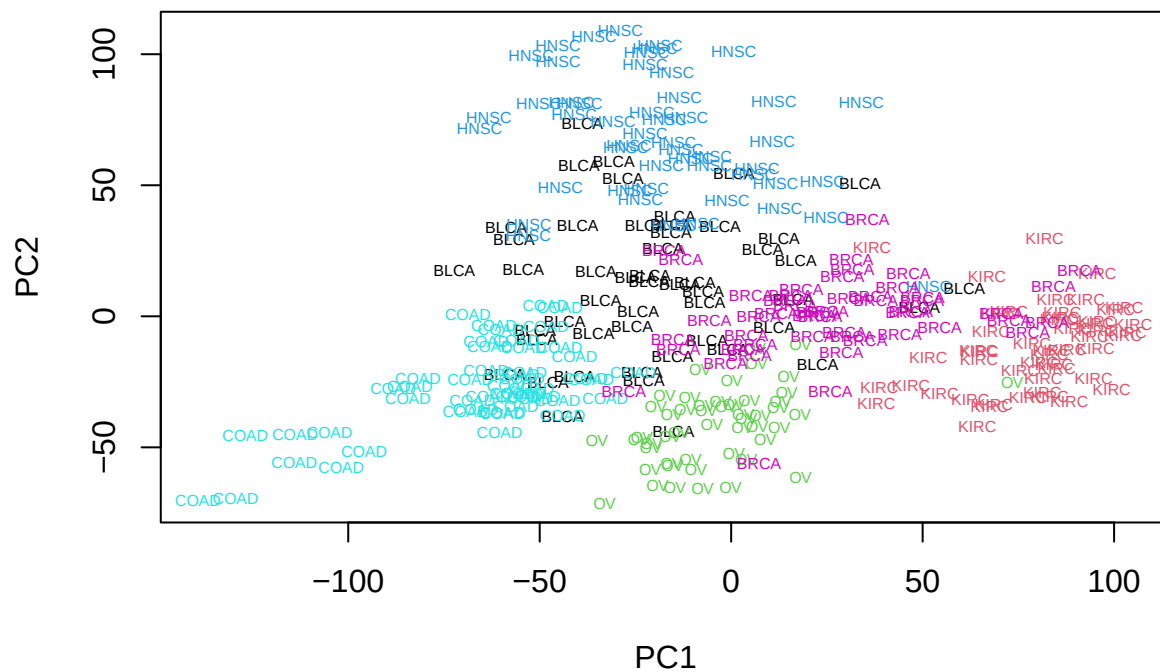


R Exercise 2: Pan Cancer (continued)

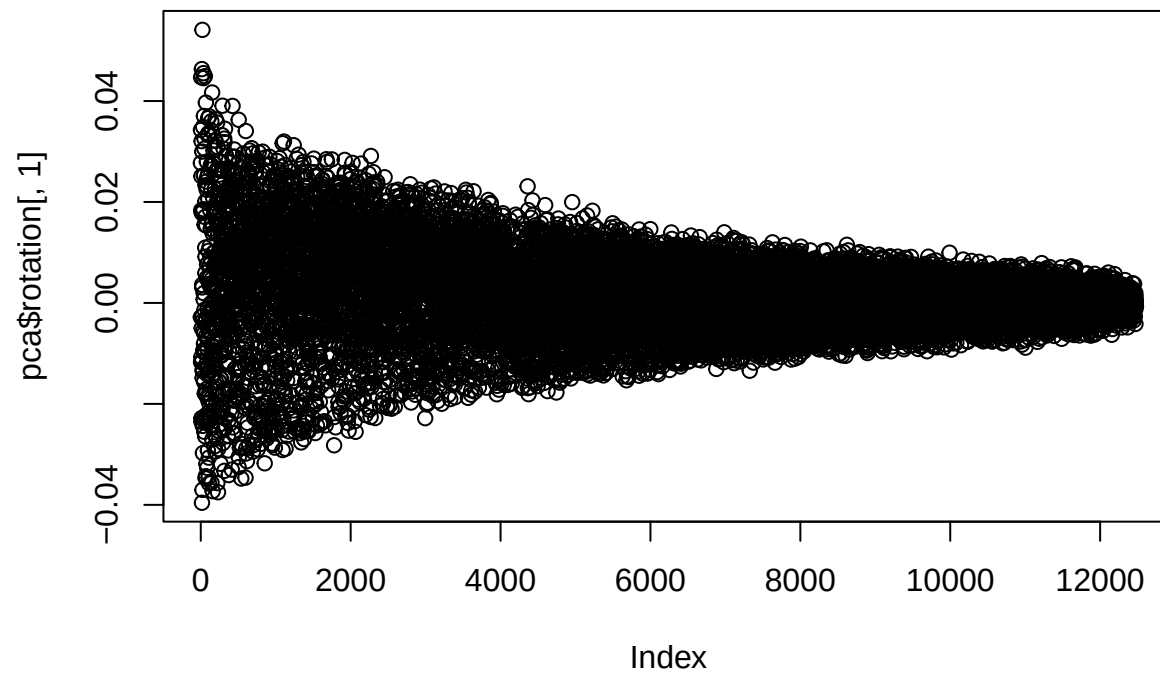
Create the first visual checks of the data or fitted objects. Use these plots to understand the structure before moving on to summaries.

Slide context: R Exercise 2: Pan Cancer (continued) What do PC1 and PC2 represent?

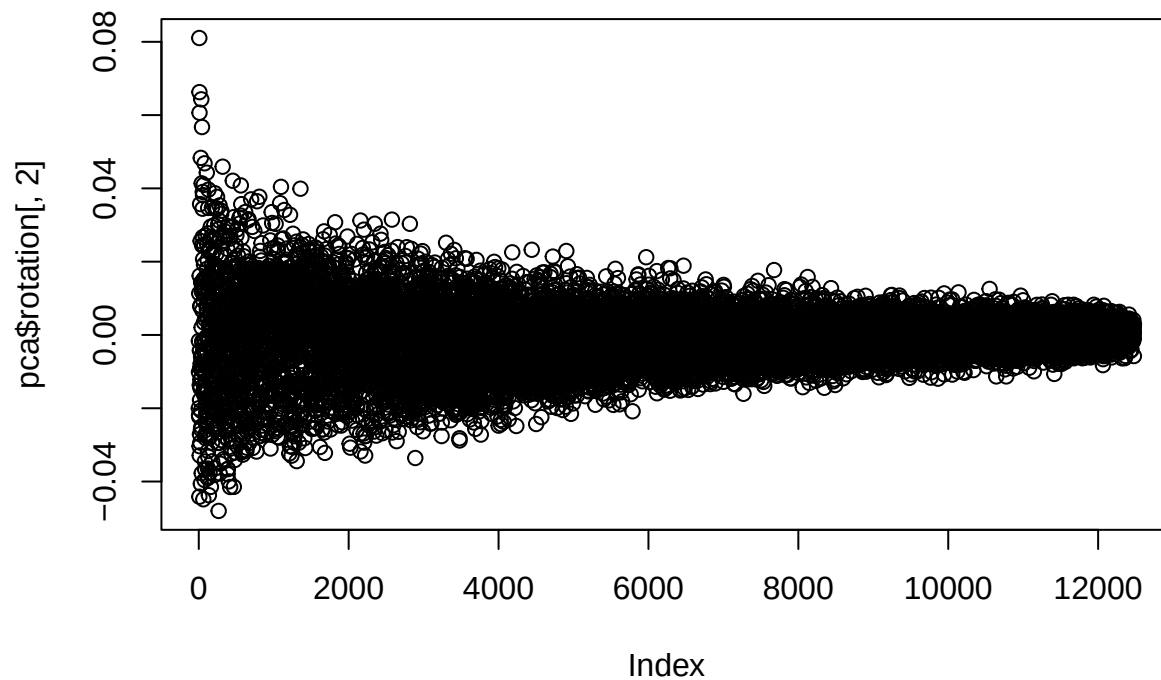
```
plot(pca$x[,1],pca$x[,2],col=gcol,xlab="PC1",ylab="PC2",type="n")
text(pca$x[,1],pca$x[,2],group,col=gcol,cex=0.5)
```



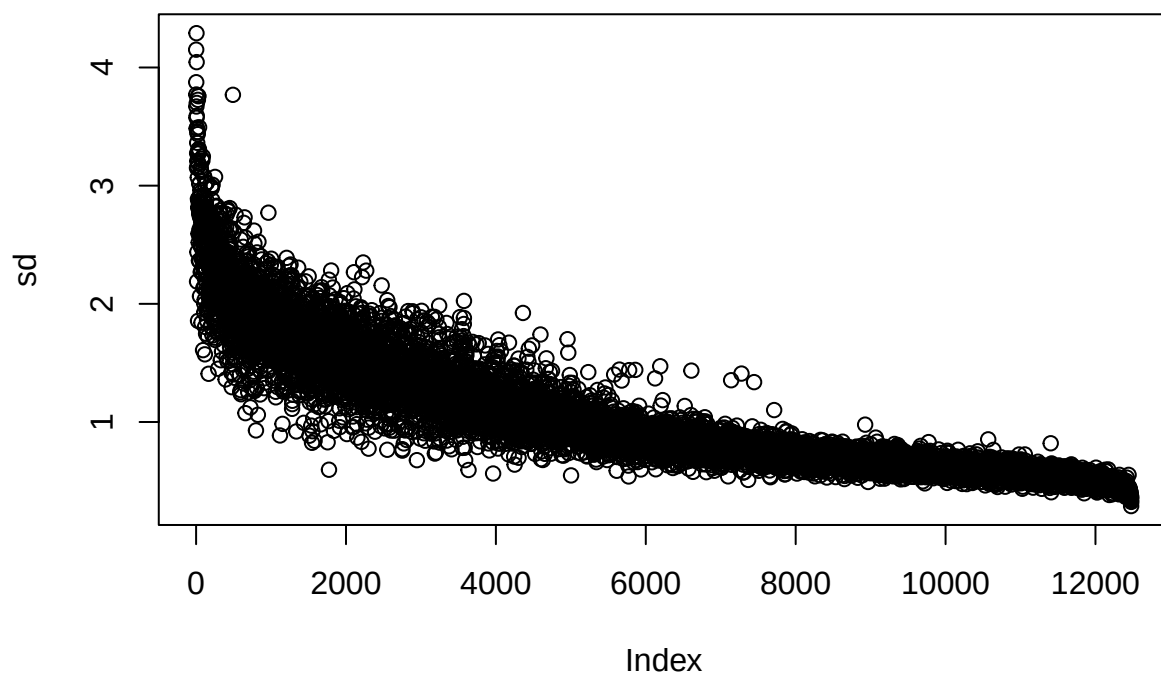
```
plot(pca$rotation[,1])
```



```
plot(pca$rotation[,2])
```

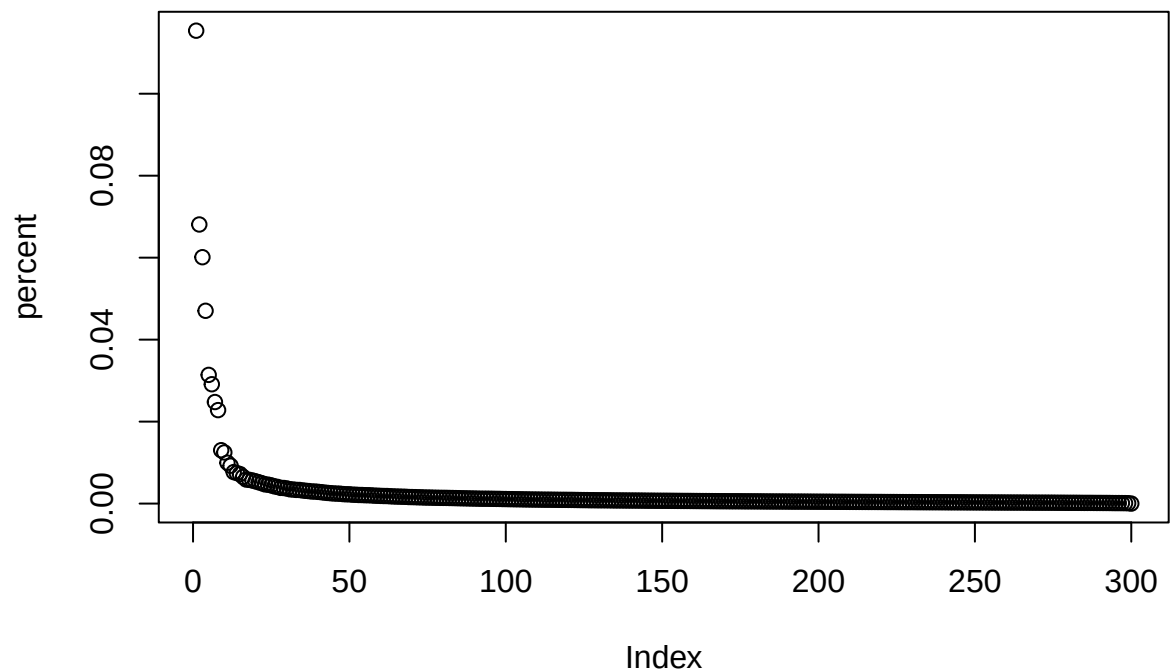
```
plot(apply(X,2,sd),ylab="sd")
```



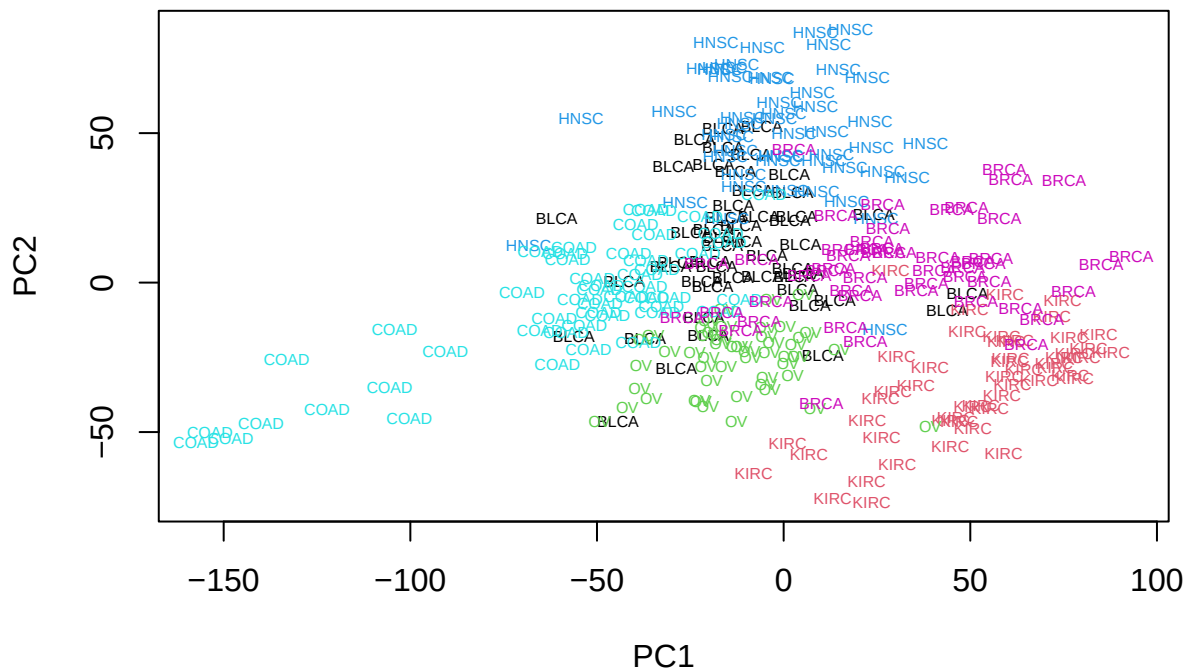
Create the first visual checks of the data or fitted objects. Use these plots to understand the structure before moving on to summaries.

Slide context: Carry out PCA with correlation matrix R

```
pca2 <- prcomp(X,scale=TRUE)
plot(pca2$sdev**2/sum(pca2$sdev**2),ylab="percent")
```



```
plot(pca2$x[,1],pca2$x[,2],col=gcol,xlab="PC1",ylab="PC2",type="n")
text(pca2$x[,1],pca2$x[,2],group,col=gcol,cex=0.5)
```



R Exercise 3: Fisher-Rao growth velocity data

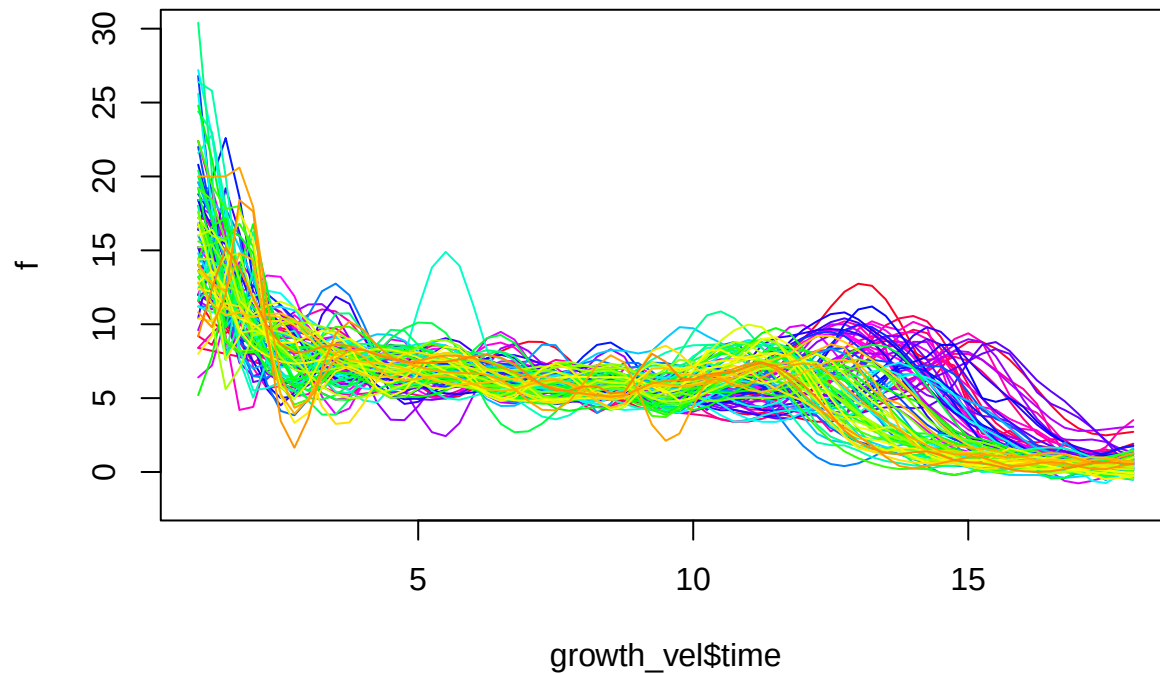
Load the data for this exercise. Check that any external files are in your working directory before evaluating the chunk.

Slide context: R Exercise 3: Fisher-Rao growth velocity data Now consider FDA with Fisher-Rao registration. Consider the functional PCA with phase-warping of the growth velocity curve data in the `fdasrvf` library.

```

library("fdastrv")
data(growth_vel)
n<-93
palette(rainbow(n+10))
plot(growth_vel$time,growth_vel$f[,1],ylim=c(-2,30),type="n",ylab="f")
for (i in 1:n){
  lines(growth_vel$time,growth_vel$f[,i],col=n+10-i)
}

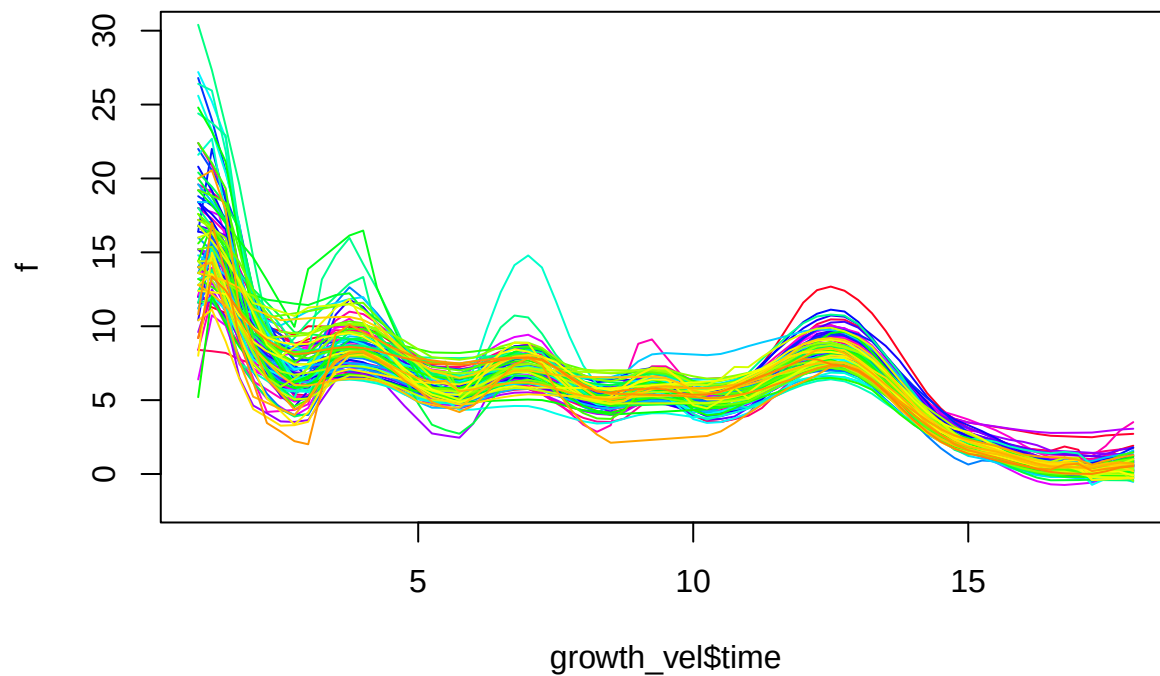
```



```

out <- time_warping( growth_vel$f, growth_vel$time)
plot(growth_vel$time,growth_vel$f[,1],ylim=c(-2,30),type="n",ylab="f")
for (i in 1:n){
  lines(growth_vel$time,out$fn[,i],col=n+10-i)
}

```

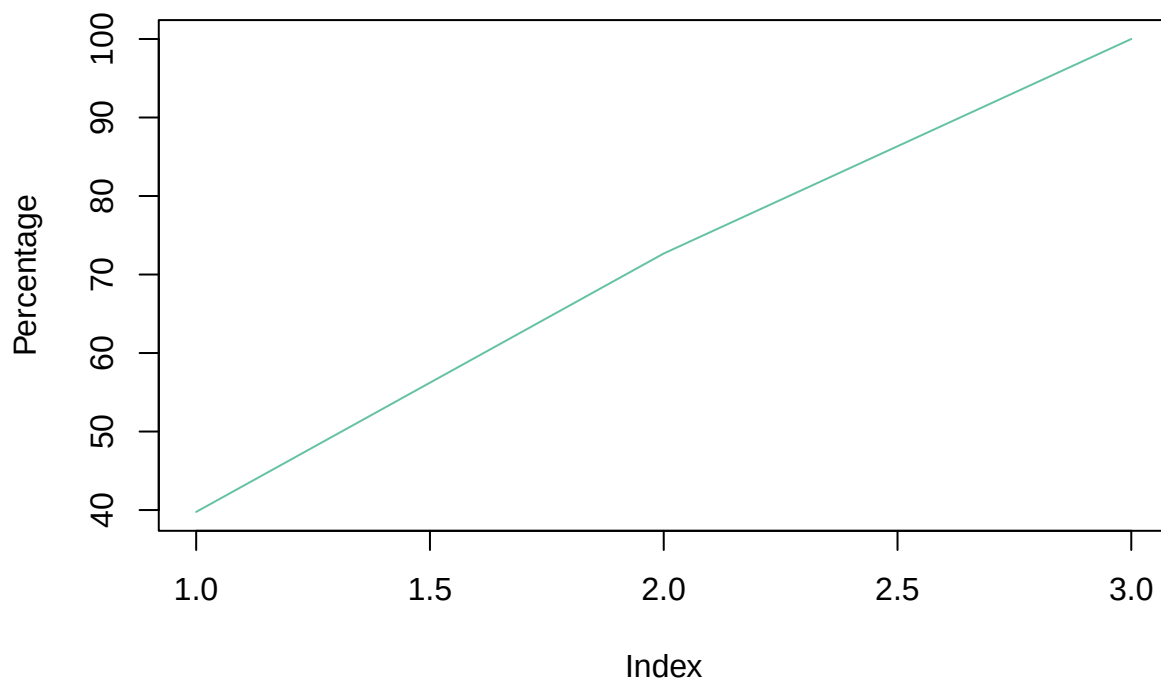
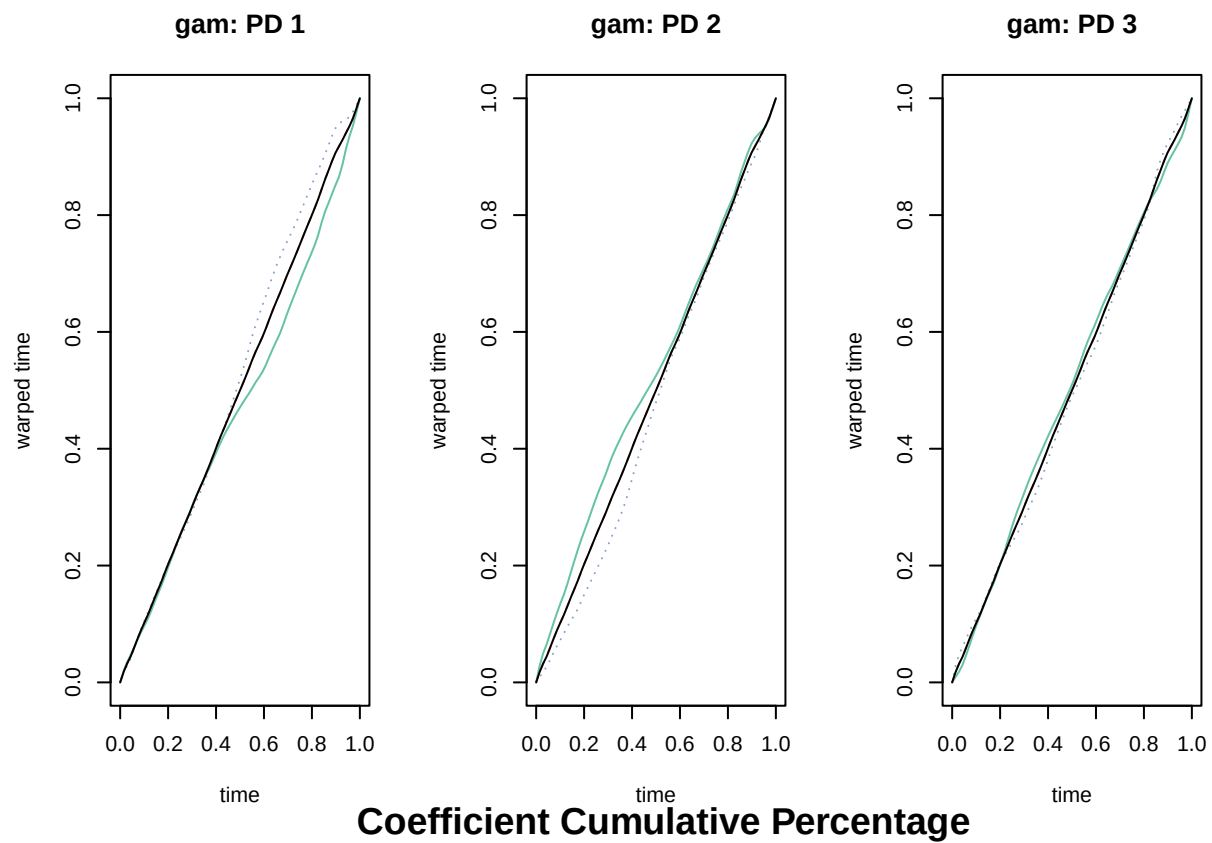


R Exercise 3: Fisher-Rao growth velocity (continued)

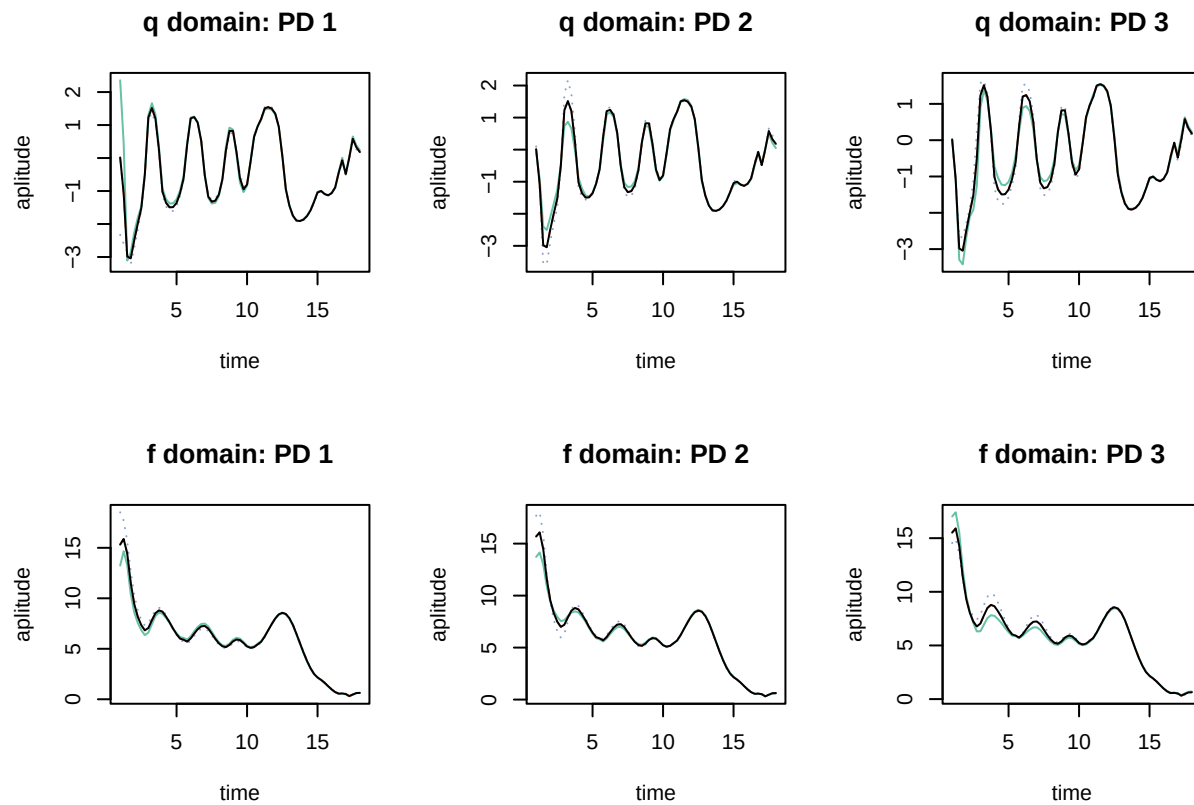
Carry out a principal component analysis and inspect the dominant modes of variation.

Slide context: R Exercise 3: Fisher-Rao growth velocity (continued) Three types of PCA can be explored:\ horizontal: phase/warp PCA\ vertical: amplitude PCA\ joint: both phase and amplitude PCA\ Fisher-Rao FPCA

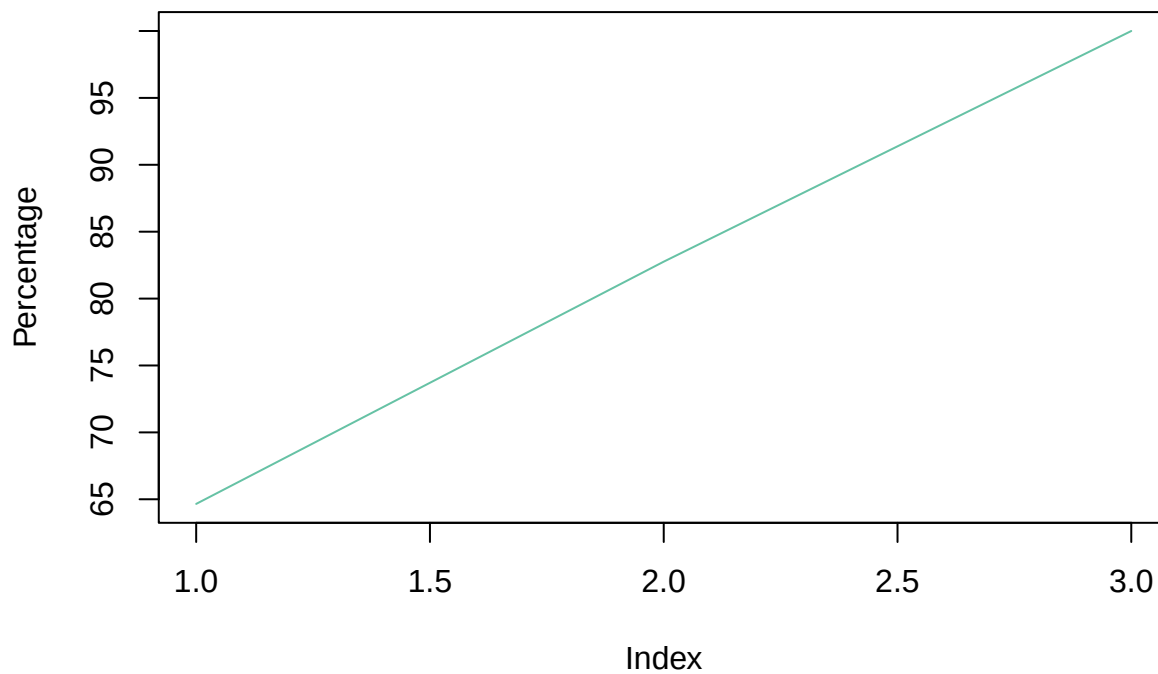
```
hfpca <- horizFPCA(out, no = 3)
```



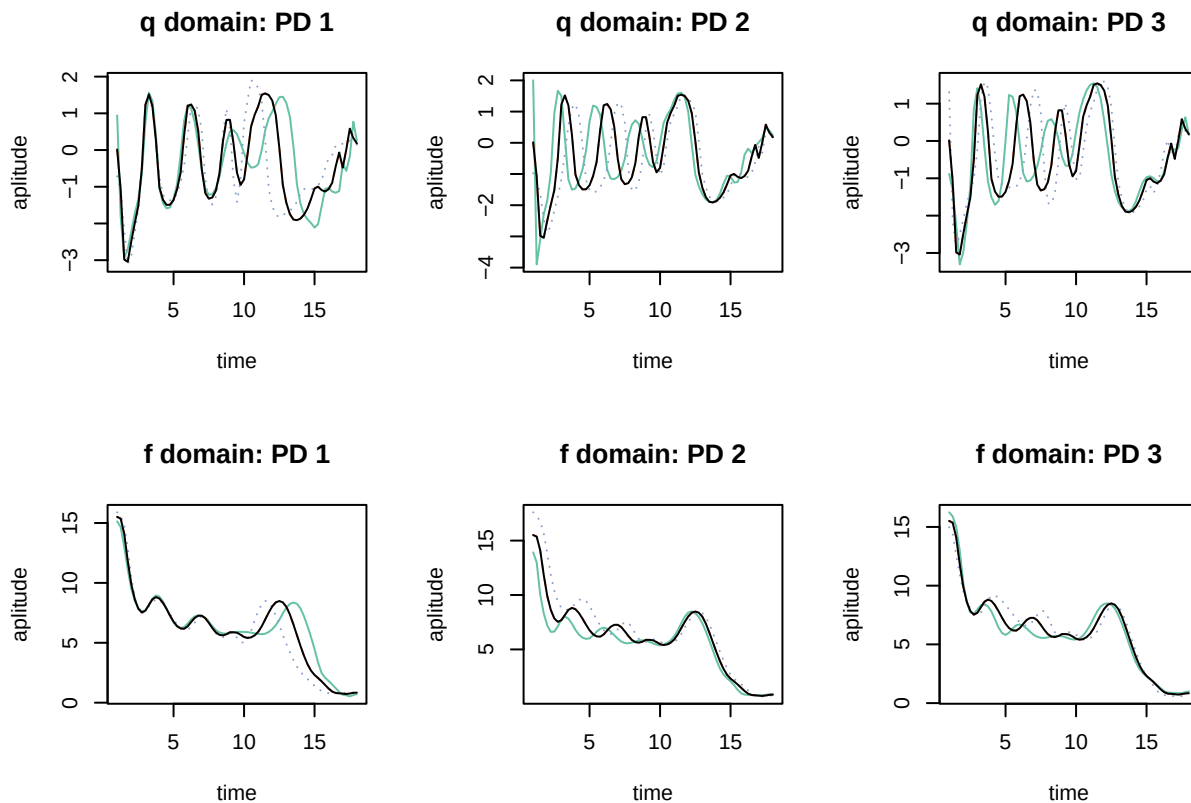
```
vfpc <- vertFPCA(out, no = 3)
```



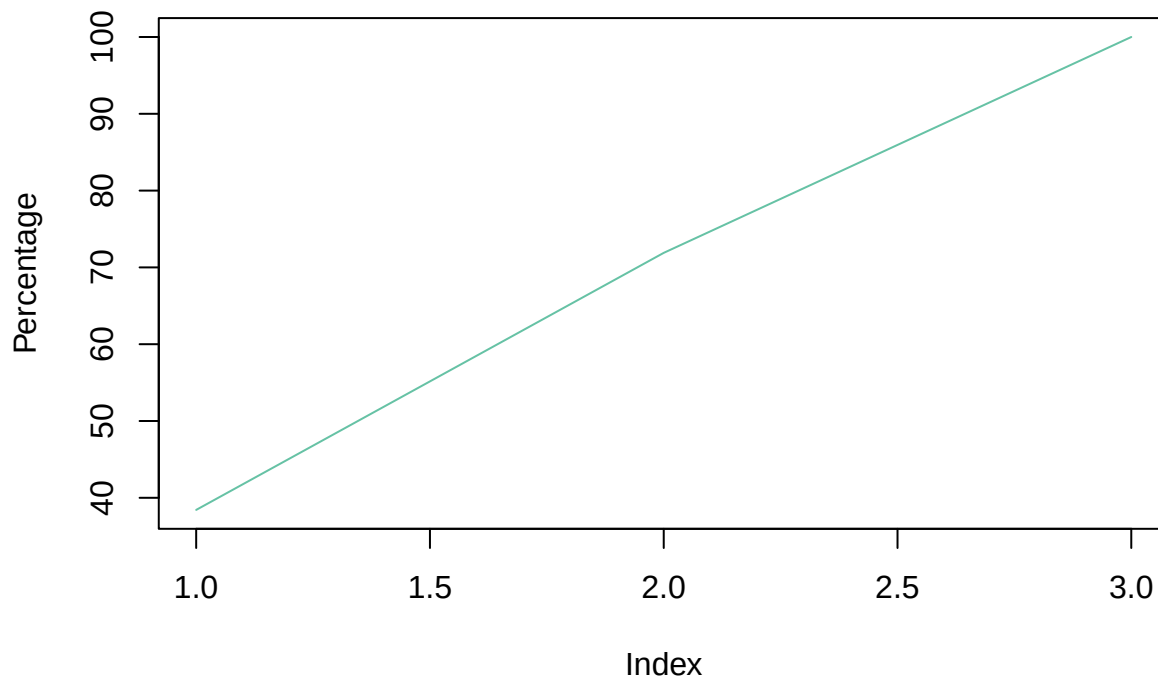
Coefficient Cumulative Percentage



```
jfpca <- jointFPCA(out, no = 3)
```



Coefficient Cumulative Percentage



R Exercise 4: novels as networks, top 100 words

Load the data for this exercise. Check that any external files are in your working directory before evaluating the chunk.

Slide context: R Exercise 4: novels as networks, top 100 words 16 Charles Dickens novels (1:16) and 7 Jane Austen novels (17:23). Select the top 100 words. Graph Laplacian

```
load("Austen-Dickens.RData")
library(GGally) ; library(network) ; library(sna) ; library(r.jive) ;
library(shapes) ; library(intergraph) ; library(igraph)
range<-1:23
nr<-max(range)
novels <- 1:nr
ng1<-1
ng2<-100
ng<-ng2-ng1+1
L <- array(0, c(ng,ng,nr) )
words<-lookup[ng1:ng2]
words
```

```
## [1] "the"      "and"      "to"       "of"       "a"        "i"        "in"
## [8] "that"     "it"       "he"       "his"      "was"      "you"      "with"
## [15] "her"      "as"       "had"      "for"      "at"       "mr"       "not"
## [22] "be"       "she"      "said"     "have"     "my"       "but"      "him"
## [29] "on"       "is"       "me"       "so"       "by"       "this"     "all"
## [36] "which"    "no"       "if"       "were"     "they"     "from"     "when"
## [43] "very"     "what"     "there"    "or"       "been"     "would"    "one"
## [50] "an"       "do"       "out"      "who"      "little"   "your"     "could"
## [57] "upon"     "mrs"      "up"       "are"      "them"     "more"     "we"
## [64] "now"      "into"     "sir"      "know"     "will"     "their"    "man"
## [71] "any"      "time"     "some"     "am"       "such"     "than"     "much"
## [78] "old"      "miss"     "then"     "say"      "never"    "again"    "should"
## [85] "before"   "about"    "down"     "well"     "being"    "good"     "how"
## [92] "has"      "here"     "come"     "did"      "like"     "himself"  "other"
## [99] "see"      "dear"
```

R Exercise 4: obtain graph Laplacians matrix and plot networks

Create the first visual checks of the data or fitted objects. Use these plots to understand the structure before moving on to summaries.

Slide context: R Exercise 4: obtain graph Laplacians matrix and plot networks

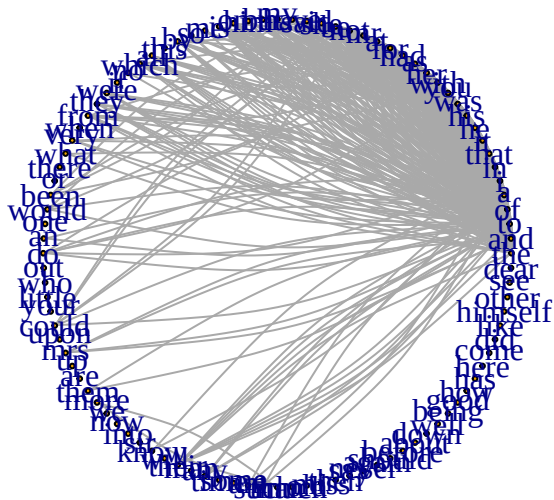
```
for (ii in 1:nr){
  print(ii)
  tem<- as.matrix( gl_wght_list[[ii]][ng1:ng2,ng1:ng2])
  A<- diag(diag(tem)) -tem #adjacency
  ones<-rep(1,times=ng)
  nodes<- c(A%%ones)
  tot<-sum(nodes)
  GL<- (diag(nodes)-A)/tot #scaled Graph Laplacian
  q90<-quantile(c(A),0.95)
  A[A<q90]<-0
  L[, ,ii]<-GL
  ggn<-network(diag(A)-A)
}
```

```
## [1] 1
## [1] 2
## [1] 3
```



```
## [1] 4
## [1] 5
## [1] 6
## [1] 7
## [1] 8
## [1] 9
## [1] 10
## [1] 11
## [1] 12
## [1] 13
## [1] 14
## [1] 15
## [1] 16
## [1] 17
## [1] 18
## [1] 19
## [1] 20
## [1] 21
## [1] 22
## [1] 23
```

```
In<-asIgraph( ggn ) #plot the last network
l2<-layout_in_circle(In)
l <- layout_with_fr(In)
plot(In,vertex.label=words,layout=l2,edge.arrow.mode=0,vertex.size=2,edge.curved=0.2)
```



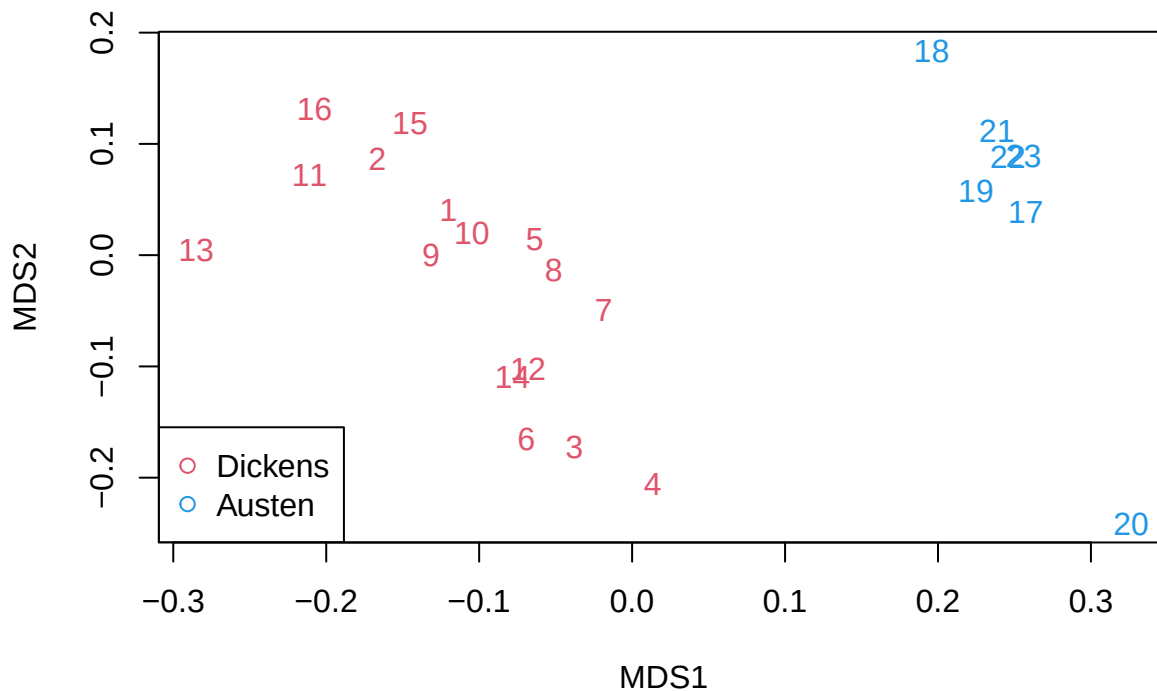
```
plot(ggnet2(ggn,edge.alpha=0.5,label=words))
```



```

for (j in (2:nr)){
  dmat[i,j]<- distcov( L[,i], L[,j] , method="Power", alpha=1/2)
  dmat[j,i]<-dmat[i,j]
}
}
palette("default")
gpc<-c(rep(2,times=16),rep(4,times=7))
mds <- cmdscale(as.dist(dmat))
plot( mds , xlab="MDS1",ylab="MDS2", type="n")
text( mds, labels=novels , col=gpc)
legend("bottomleft",legend=c("Dickens","Austen"),pch=c(1,1),col=c(2,4))

```



R Exercise 5: Pairwise Procrustes Fit

Compute distances or an ordination, then use the output to represent relationships between observations.

Slide context: R Exercise 5: Pairwise Procrustes Fit

```

centre <- function(X) scale(X, scale = FALSE)
csize <- function(X) sqrt(sum(centre(X)^2))
preshape <- function(X) centre(X) / csize(X)
X <- dna.dat[,1]
Y <- dna.dat[,30]
X0 <- preshape(X)
Y0 <- preshape(Y)
sv <- svd(t(X0) %*% Y0)
R <- sv$v %*% t(sv$u)
R

```

```

##           [,1]      [,2]      [,3]
## [1,]  0.99937670  0.03191854 0.01508071
## [2,] -0.03208470  0.99942563 0.01090805

```

```
## [3,] -0.01472388 -0.01138511 0.99982678
Y_aligned <- Y0 %*% R
dP <- sqrt(sum((X0 - Y_aligned)^2))
dP

## [1] 0.1142859
print( sqrt( 2*(1-cos( riemdist( X,Y) ) ))) #check

## [1] 0.1142859
print( riemdist( X,Y) ) # similar

## [1] 0.1143482
```

R Exercise 5: Pairwise Procrustes Fit (continued)

Run this chunk as the next step in the exercise. The code is kept close to the original course material, with explanatory text added around it.

Slide context: R Exercise 5: Pairwise Procrustes Fit (continued) Carry out ordinary Procrustes registration on X,Y, with and without scaling.

```
X <- dna.dat[,1]
Y <- dna.dat[,30]
outline=c(1:22)
shapes3d( abind( X, Y) , joinline=outline )

#match Y to X, without scaling
opa1 <- procOPA( X, Y, scale=FALSE)
opa1
```

```
## $R
##           [,1]      [,2]      [,3]
## [1,]  0.99937670  0.03191854  0.01508071
## [2,] -0.03208470  0.99942563  0.01090805
## [3,] -0.01472388 -0.01138511  0.99982678
##
## $s
## [1] 1
##
## $Ahat
##           [,1]      [,2]      [,3]
## [1,]  8.1828636 -6.3862727 -12.8293636
## [2,] 10.0998636  0.5157273 -12.6063636
## [3,]  7.1418636  6.7457273 -10.3863636
## [4,]  1.7758636 10.4037273  -8.5223636
## [5,] -3.1741364 10.6767273  -3.9553636
## [6,] -7.2031364  7.7447273   0.6106364
## [7,] -10.0361364 2.6797273   4.0626364
## [8,] -9.4421364 -3.4632727   7.2806364
## [9,] -7.0571364 -6.3732727  12.4486364
## [10,] -1.1971364 -9.5042727  14.0586364
## [11,]  4.9958636 -7.5262727  16.6456364
## [12,]  6.0738636  9.4297273  18.6706364
## [13,]  9.0198636  5.3577273  14.9356364
```

```
## [14,] 10.1528636 -0.4222727 11.2446364
## [15,] 8.4878636 -5.7212727 7.2796364
## [16,] 4.5978636 -8.0272727 2.3906364
## [17,] -0.8841364 -10.6282727 -1.8283636
## [18,] -6.5161364 -8.0392727 -4.3573636
## [19,] -8.1281364 -4.2642727 -9.2113636
## [20,] -10.0061364 1.6537273 -12.0353636
## [21,] -6.4611364 6.1837273 -15.8563636
## [22,] -0.4231364 8.9647273 -18.0393636
##
## $Bhat
##      [,1]      [,2]      [,3]
## [1,] 9.6511879 -4.621424 -13.8123141
## [2,] 10.7857673 2.109556 -12.1356854
## [3,] 7.2646333 7.267736 -9.8614097
## [4,] 1.3626211 10.000723 -7.7328262
## [5,] -3.3845841 9.654665 -4.1990513
## [6,] -8.2755472 6.656452 -0.1134925
## [7,] -10.0353296 2.516379 4.5832751
## [8,] -10.1141932 -2.879858 8.7644014
## [9,] -6.5454768 -6.431991 12.6691817
## [10,] -0.9082736 -9.211045 14.9909456
## [11,] 3.7290591 -7.111613 18.3427080
## [12,] 6.7390908 10.373680 15.8056599
## [13,] 9.6716320 4.894318 12.8999518
## [14,] 10.2524691 -1.085556 10.3399790
## [15,] 8.0319673 -6.901412 7.3395507
## [16,] 3.4969724 -10.322990 4.0492530
## [17,] -2.4089712 -11.454103 -0.1703142
## [18,] -4.5378290 -8.479470 -5.9667968
## [19,] -7.9391749 -4.110074 -10.0938467
## [20,] -9.9637163 2.596581 -11.5385501
## [21,] -6.4808041 7.138010 -15.7622950
## [22,] -0.3915007 9.401438 -18.3983241
##
## $OSS
## [1] 66.39779
##
## $rmsd
## [1] 1.737263

shapes3d( abind( opa1$Ahat , opa1$Bhat) , joinline=outline )

#match Y to X, with scaling
opa2 <- procOPA( X, Y, scale=TRUE)
opa2

## $R
##      [,1]      [,2]      [,3]
## [1,] 0.99937670 0.03191854 0.01508071
## [2,] -0.03208470 0.99942563 0.01090805
## [3,] -0.01472388 -0.01138511 0.99982678
##
## $s
## [1] 0.9865723
```

```

##
## $Ahat
##      [,1]      [,2]      [,3]
## [1,]  8.1828636 -6.3862727 -12.8293636
## [2,] 10.0998636  0.5157273 -12.6063636
## [3,]  7.1418636  6.7457273 -10.3863636
## [4,]  1.7758636 10.4037273 -8.5223636
## [5,] -3.1741364 10.6767273 -3.9553636
## [6,] -7.2031364  7.7447273  0.6106364
## [7,] -10.0361364 2.6797273  4.0626364
## [8,] -9.4421364 -3.4632727  7.2806364
## [9,] -7.0571364 -6.3732727 12.4486364
## [10,] -1.1971364 -9.5042727 14.0586364
## [11,]  4.9958636 -7.5262727 16.6456364
## [12,]  6.0738636  9.4297273 18.6706364
## [13,]  9.0198636  5.3577273 14.9356364
## [14,] 10.1528636 -0.4222727 11.2446364
## [15,]  8.4878636 -5.7212727  7.2796364
## [16,]  4.5978636 -8.0272727  2.3906364
## [17,] -0.8841364 -10.6282727 -1.8283636
## [18,] -6.5161364 -8.0392727 -4.3573636
## [19,] -8.1281364 -4.2642727 -9.2113636
## [20,] -10.0061364  1.6537273 -12.0353636
## [21,] -6.4611364  6.1837273 -15.8563636
## [22,] -0.4231364  8.9647273 -18.0393636
##
## $Bhat
##      [,1]      [,2]      [,3]
## [1,]  9.5215951 -4.559370 -13.6268472
## [2,] 10.6409398  2.081229 -11.9727316
## [3,]  7.1670864  7.170147 -9.7289941
## [4,]  1.3443243  9.866436 -7.6289925
## [5,] -3.3391371  9.525025 -4.1426679
## [6,] -8.1644260  6.567071 -0.1119686
## [7,] -9.9005787  2.482590  4.5217324
## [8,] -9.9783833 -2.841189  8.6467161
## [9,] -6.4575864 -6.345624 12.4990644
## [10,] -0.8960776 -9.087362 14.7896524
## [11,]  3.6789866 -7.016121 18.0964085
## [12,]  6.6486006 10.234386 15.5934270
## [13,]  9.5417647  4.828599 12.7267357
## [14,] 10.1148026 -1.070980 10.2011374
## [15,]  7.9241168 -6.808742  7.2409978
## [16,]  3.4500163 -10.184377  3.9948810
## [17,] -2.3766244 -11.300301 -0.1680273
## [18,] -4.4768966 -8.365611 -5.8866768
## [19,] -7.8325705 -4.054885 -9.9583100
## [20,] -9.8299270  2.561715 -11.3836144
## [21,] -6.3937821  7.042163 -15.5506444
## [22,] -0.3862437  9.275198 -18.1512778
##
## $OSS
## [1] 65.47822
##

```

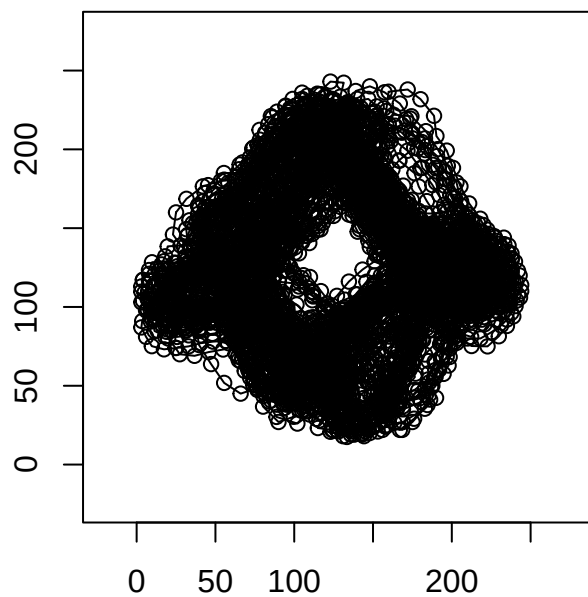
```
## $rmsd
## [1] 1.725191
shapes3d( abind( opa2$Ahat , opa2$Bhat) , joinline=outline )
```

R Exercise 6: GPA with the shapes package

Load the data for this exercise. Check that any external files are in your working directory before evaluating the chunk.

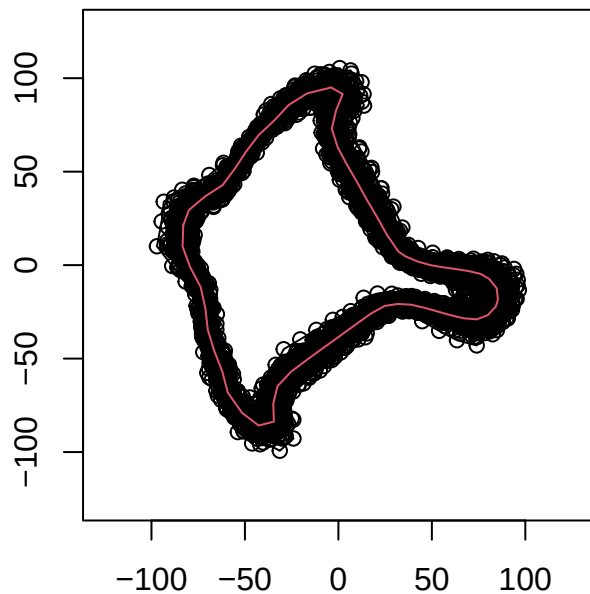
Slide context: R Exercise 6: GPA with the shapes package T2 vertebrae outlines

```
data(mice)
outline=c(1:60,1)
plotshapes(mice$outlines,joinline=outline)
```



```
# mice$outlines is k x m x n landmark array
gpa <- procGPA(mice$outlines)
aligned <- gpa$rotated. # aligned configurations
mean_shape <- gpa$mshape # mean shape

plotshapes(gpa$rotated,joinline=outline)
lines( mean_shape[c(1:60,1),], col=2)
```

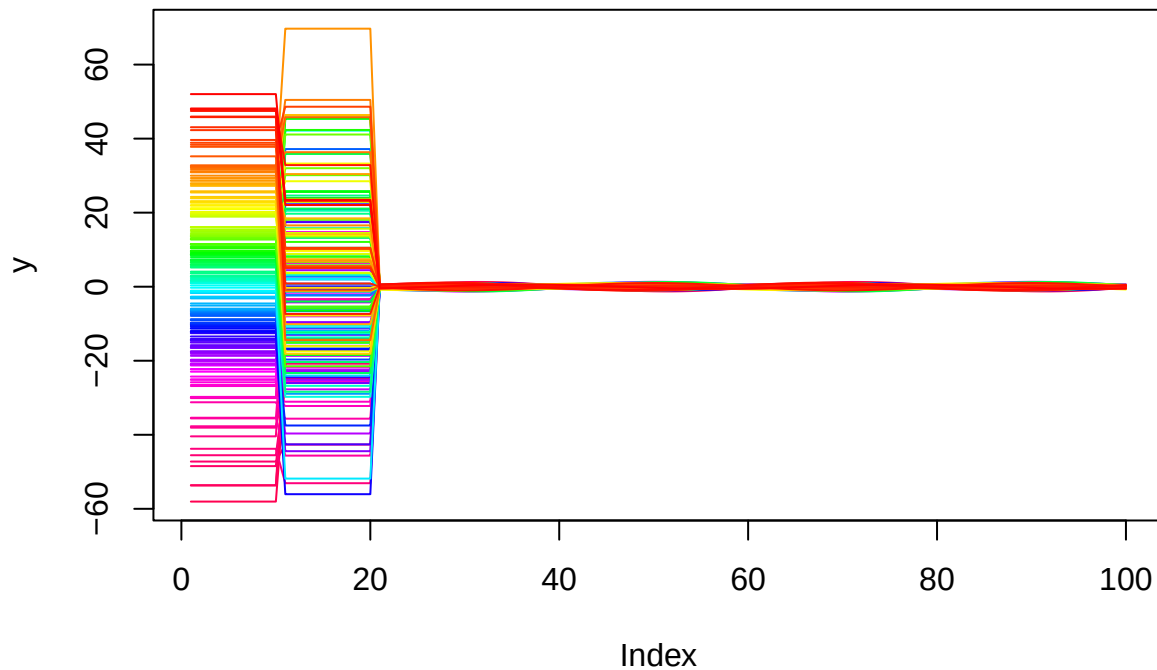


R Exercise 7: Two Scale Curves Data

Create the first visual checks of the data or fitted objects. Use these plots to understand the structure before moving on to summaries.

Slide context: R Exercise 7: Two Scale Curves Data Read in the data in TwoScaleCurvesData.xlsx and plot the curves. There are $n=200$ piecewise linear curves in $d=100$ dimensions. What structure do you see in the data objects?

```
library("readxl")
x <- read_xlsx("TwoScaleCurvesData.xlsx", col_names=FALSE)
y <- as.matrix(t(x))
n <- dim(y)[1]
y <- y[order(y[,1]),]
palette(rainbow(n+10))
plot( y[1,], type="n", ylim=c(min(y),max(y)), ylab="y")
for ( i in 1:n){
  lines(1:100,y[i,],col=n+1-i)
}
```

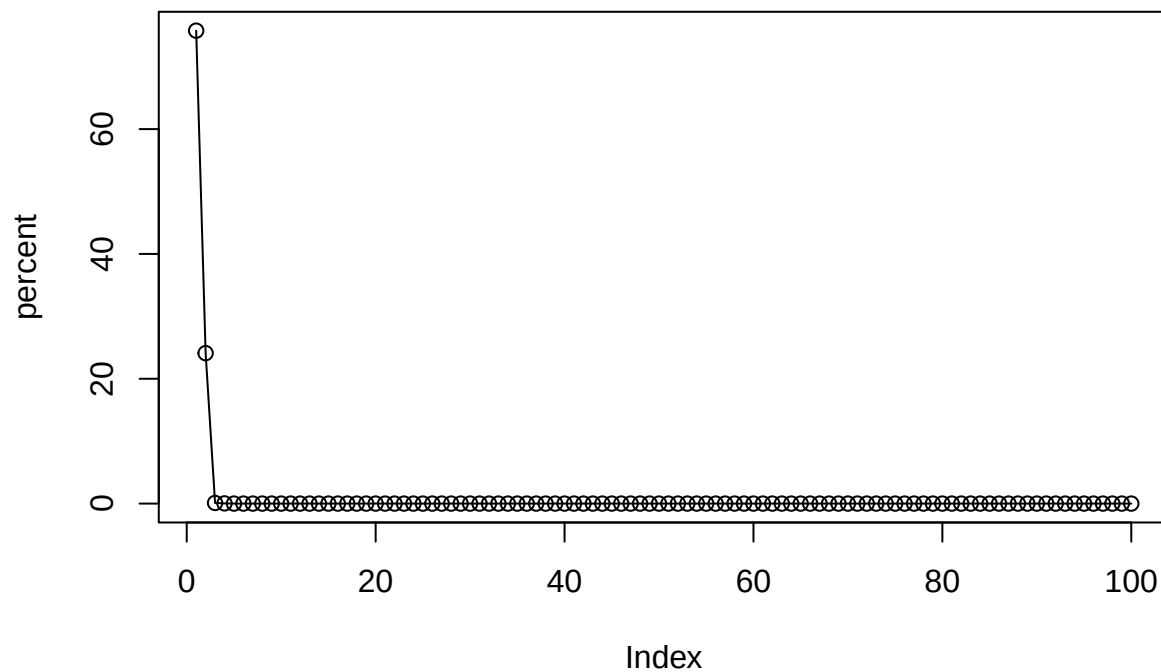



R Exercise 7: PCA: Two Scale Curves Data.

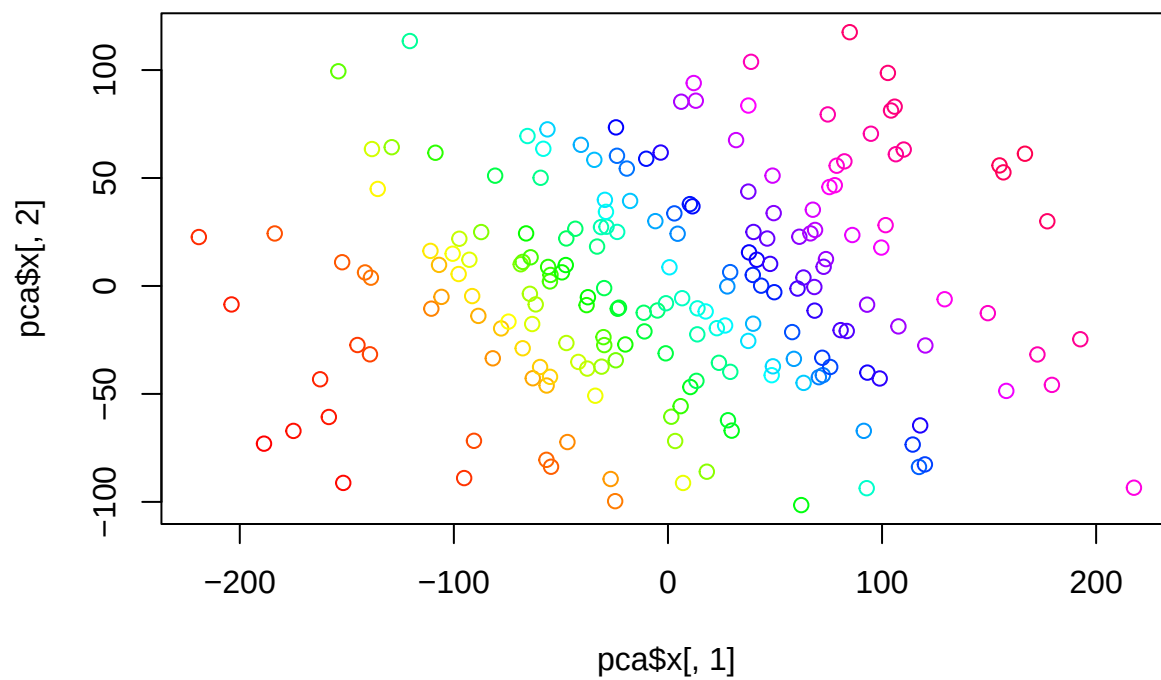
Create the first visual checks of the data or fitted objects. Use these plots to understand the structure before moving on to summaries.

Slide context: R Exercise 7: PCA: Two Scale Curves Data. Now carry out PCA of the data. In particular obtain a scree plot and plot the first two PC scores. Do you see any patterns of interest?

```
pca <- prcomp(y)
percent <- 100*pca$sdev**2/sum(pca$sdev**2)
plot(percent)
lines(percent)
```



```
#PC scores
plot( pca$x[,1], pca$x[,2], col=1:n)
```

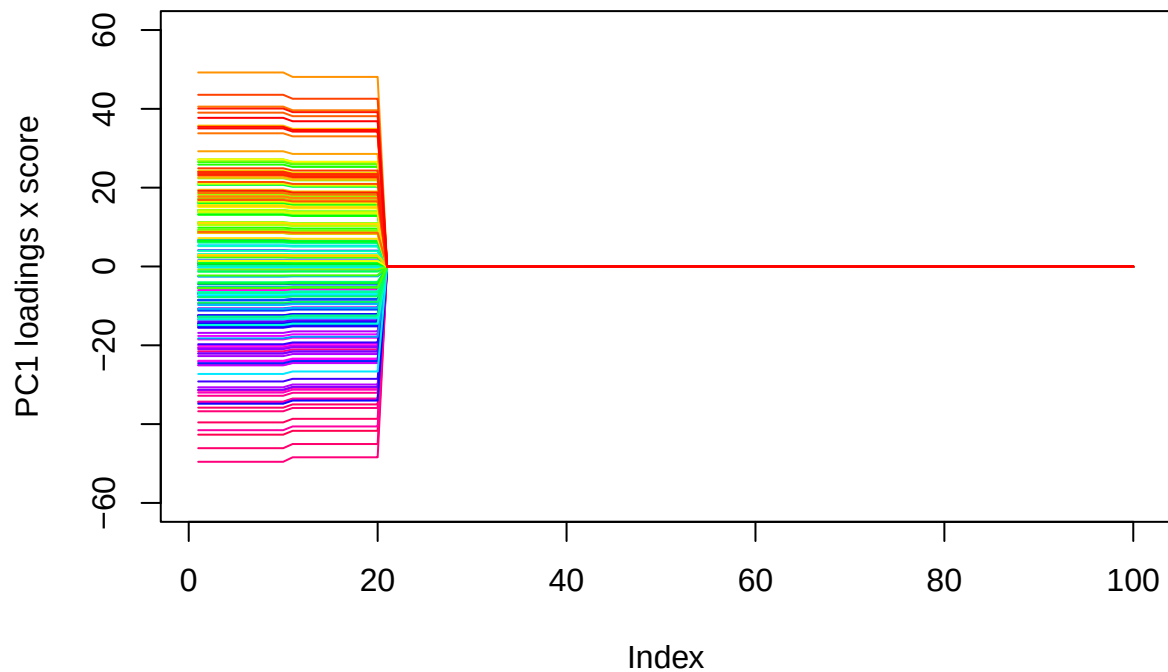


R Exercise 7: Modes of variation: Two Scale Curves Data.

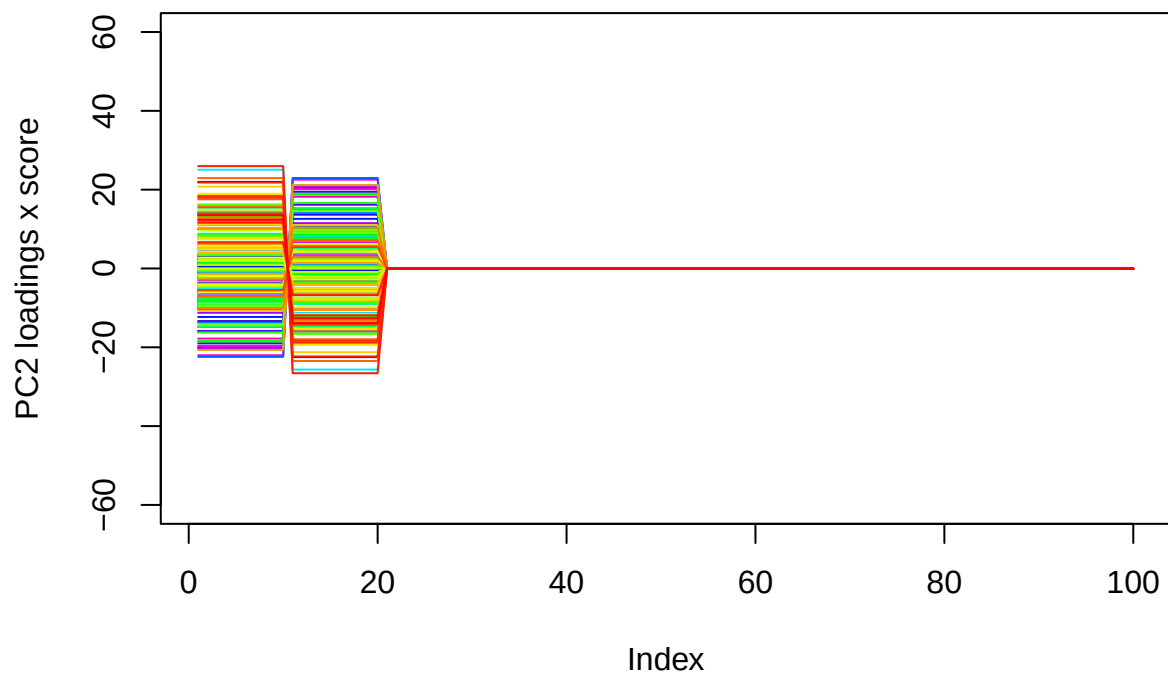
Create the first visual checks of the data or fitted objects. Use these plots to understand the structure before moving on to summaries.

Slide context: R Exercise 7: Modes of variation: Two Scale Curves Data. Investigate the first three modes of variation, and describe what you see.

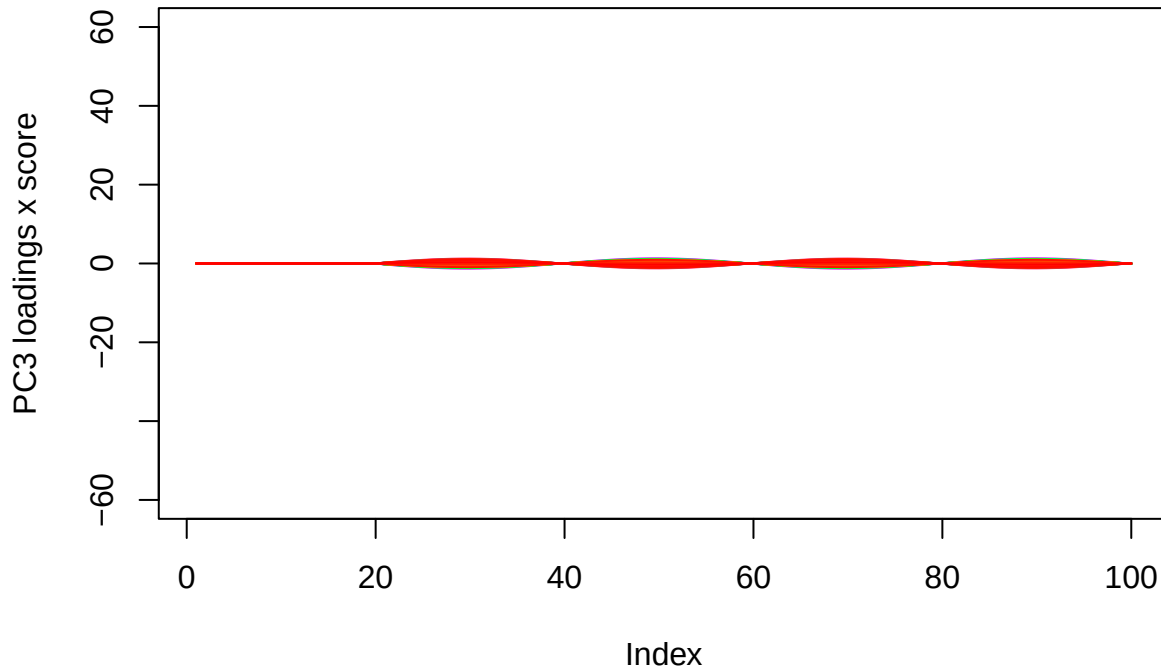
```
plot(pca$rotation[,1]*pca$x[i,1],type="n",ylab="PC1 loadings x score",ylim=c(-60,60))
for (i in 1:n){
modeofvar1 <- pca$rotation[,1]*pca$x[i,1]
  lines(1:100,modeofvar1,col=n+1-i)
}
```



```
plot(pca$rotation[,1]*pca$x[i,1],type="n",ylab="PC2 loadings x score",ylim=c(-60,60))
for (i in 1:n){
modeofvar2 <- pca$rotation[,2]*pca$x[i,2]
  lines(1:100,modeofvar2,col=n+1-i)
}
```



```
plot(pca$rotation[,1]*pca$x[i,1],type="n",ylab="PC3 loadings x score",ylim=c(-60,60))
for (i in 1:n){
  modeofvar3 <- pca$rotation[,3]*pca$x[i,3]
  lines(1:100,modeofvar3,col=n+1-i)
}
```

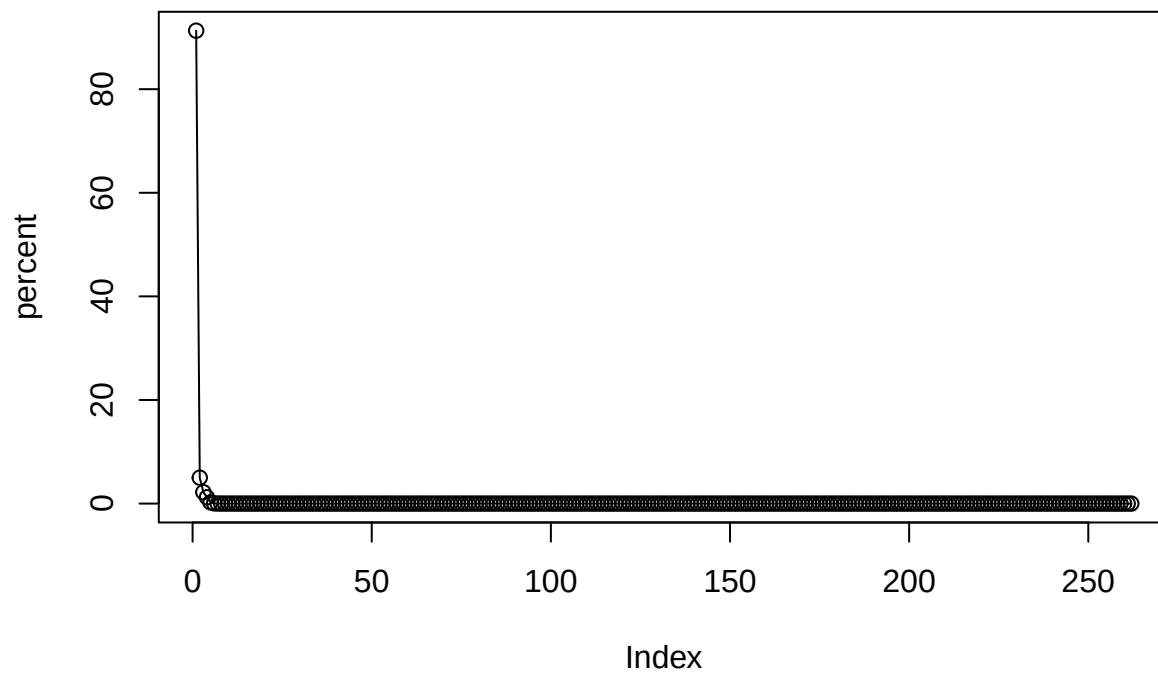


R Exercise 8: Drug Discovery Data

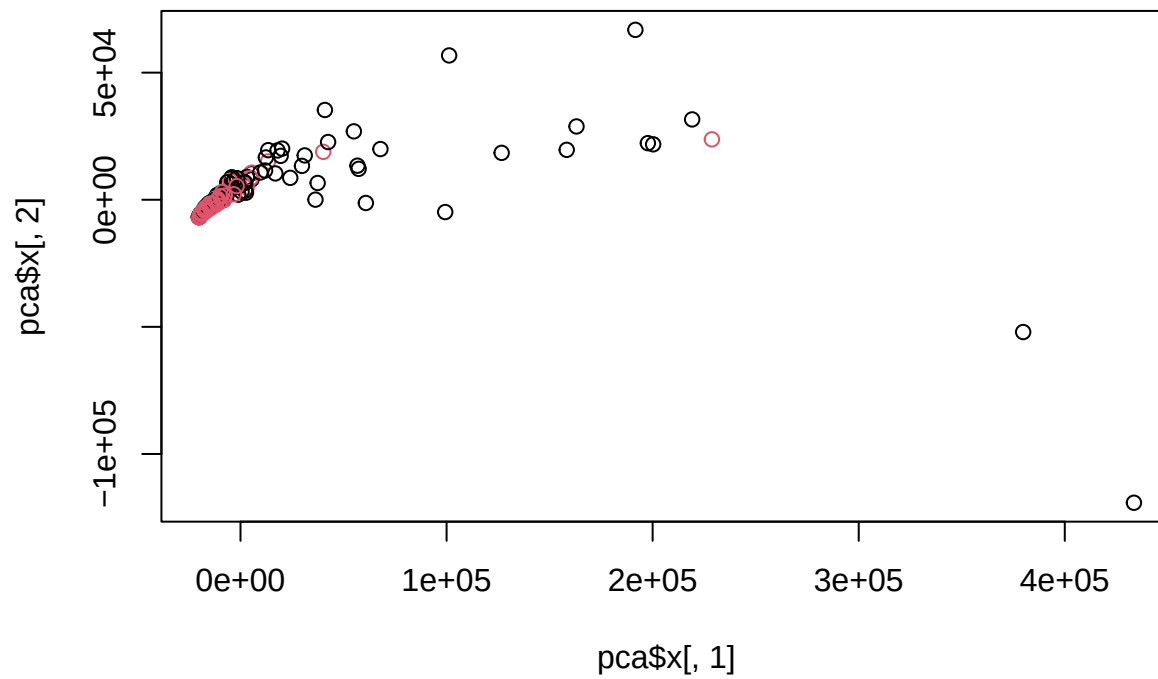
Create the first visual checks of the data or fitted objects. Use these plots to understand the structure before moving on to summaries.

Slide context: R Exercise 8: Drug Discovery Data Read in the Drug Discovery Data. There are two groups given in the first column of the spreadsheet. Carry out PCA.

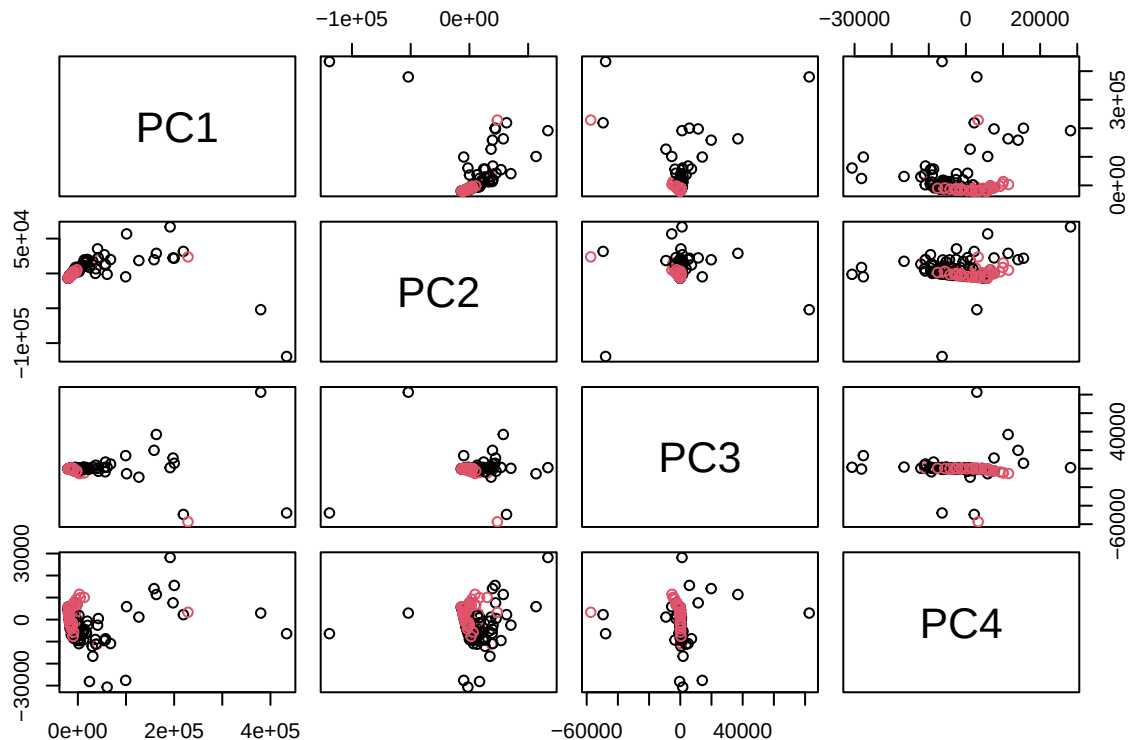
```
palette("default")
x <- read_xlsx("DrugDiscoveryData.xlsx")
x <- as.matrix(x)
group <- as.double(x[,1])+1
xn <- as.matrix(x[,c(-1,-2,-3)])
n<-dim(xn)[1]
pca<-prcomp(xn)
percent <- 100*pca$sdev**2/sum(pca$sdev**2)
plot(percent)
lines(percent)
```



```
plot( pca$x[,1], pca$x[,2], col=group)
```



```
pairs( pca$x[,1:4] ,col=group)
```



R Exercise 8: Standardizing: Drug Discovery Data

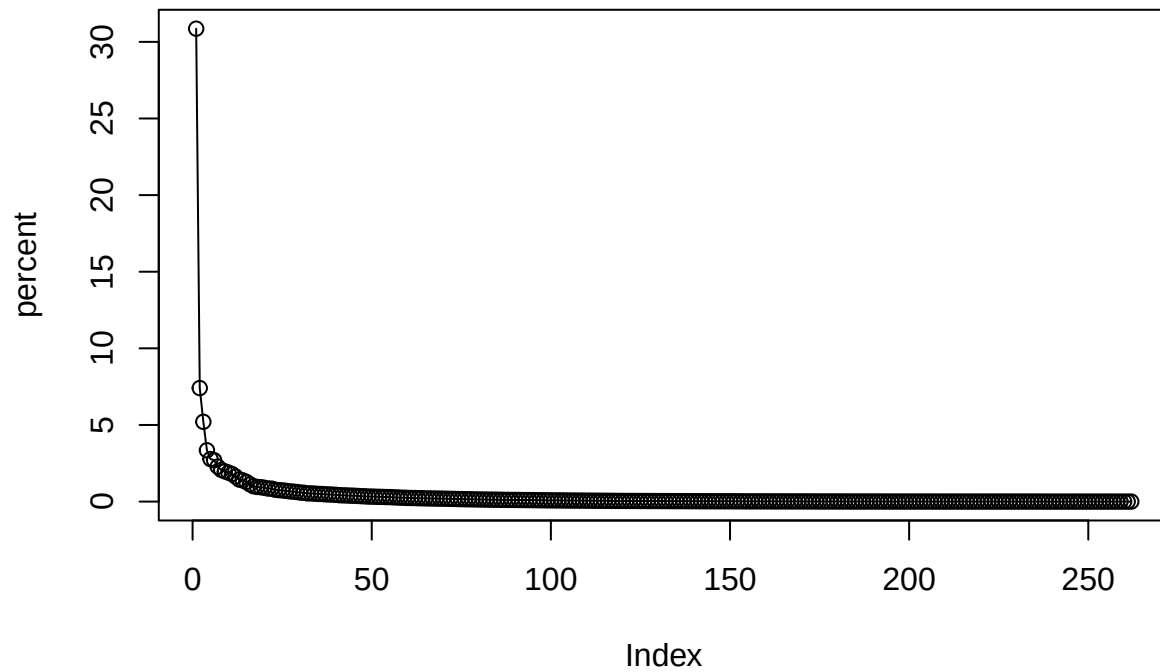
Create the first visual checks of the data or fitted objects. Use these plots to understand the structure before moving on to summaries.

Slide context: R Exercise 8: Standardizing: Drug Discovery Data Do you think the data should be standardized here? We now try standardizing, but we need to remove variables with no variability.

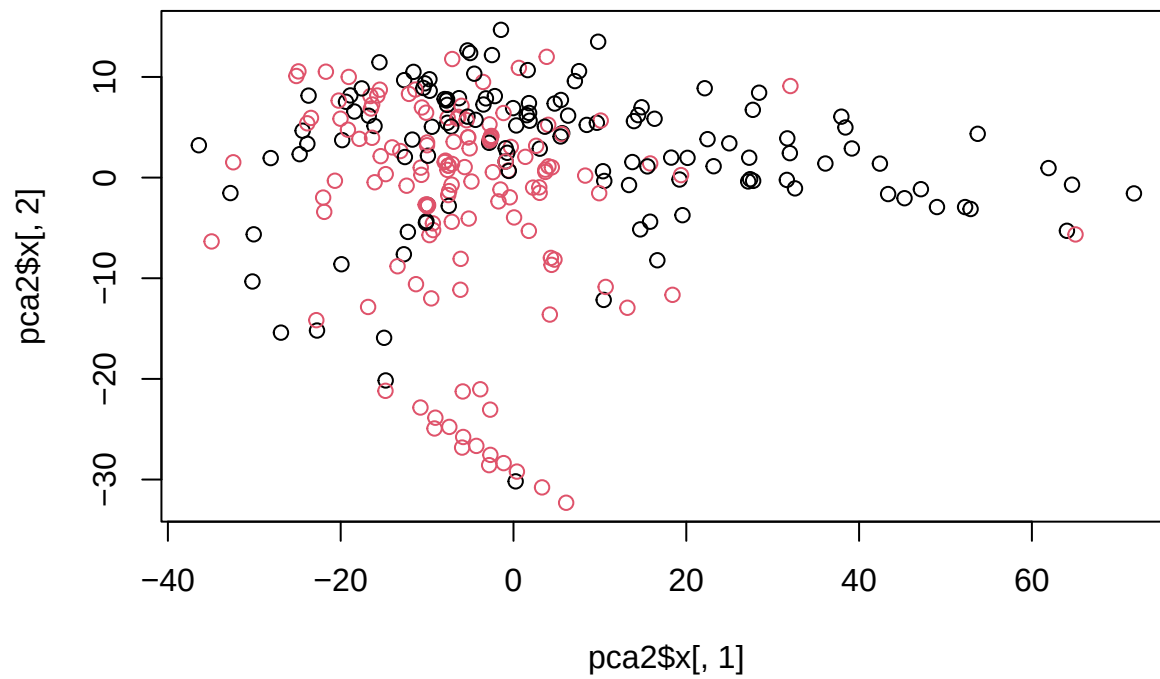
```
sdev <- apply( xn , 2, sd)
xn2 <- xn[,sdev!=0]
sd2 <- sdev[sdev!=0]
center <- apply( xn2, 2, mean)
dim(xn2)

## [1] 262 1174

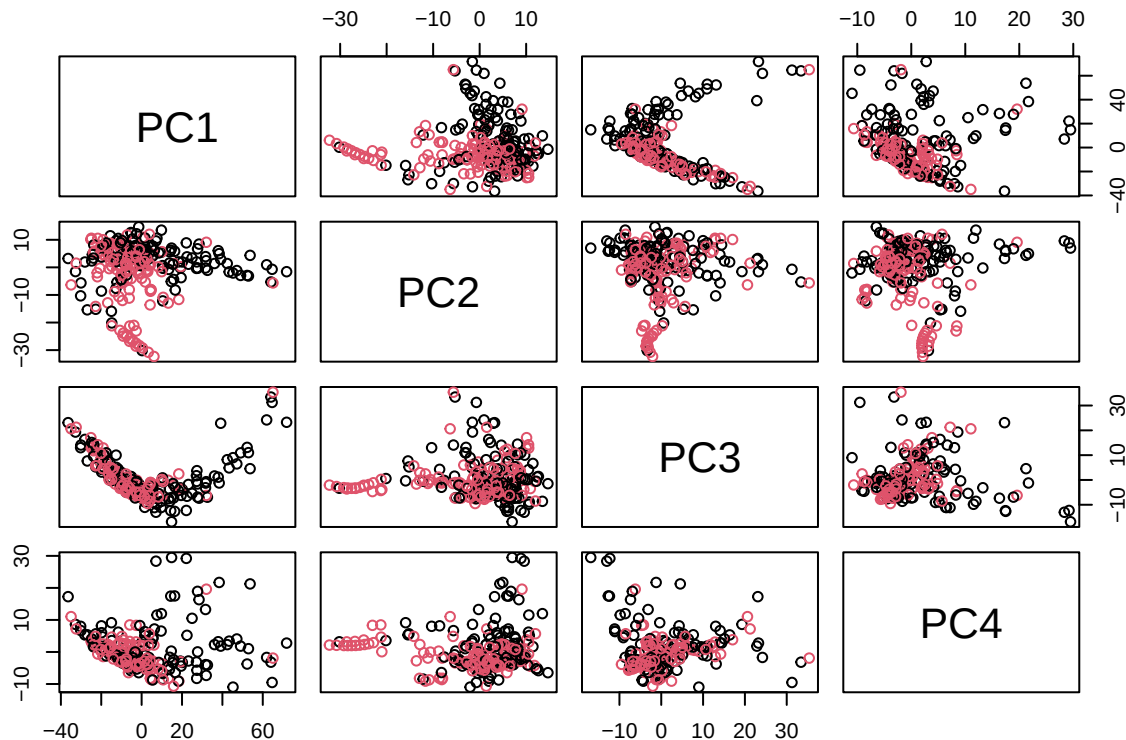
ones<- rep(1,times=n)
xn3 <- (xn2 - ones%*%t(center)) %*%diag(1/sd2)
pca2<-prcomp(xn3,scale=TRUE)
percent <- 100*pca2$sdev**2/sum(pca2$sdev**2)
plot(percent)
lines(percent)
```



```
plot( pca2$x[,1], pca2$x[,2], col=group)
```



```
pairs( pca2$x[,1:4],col=group)
```

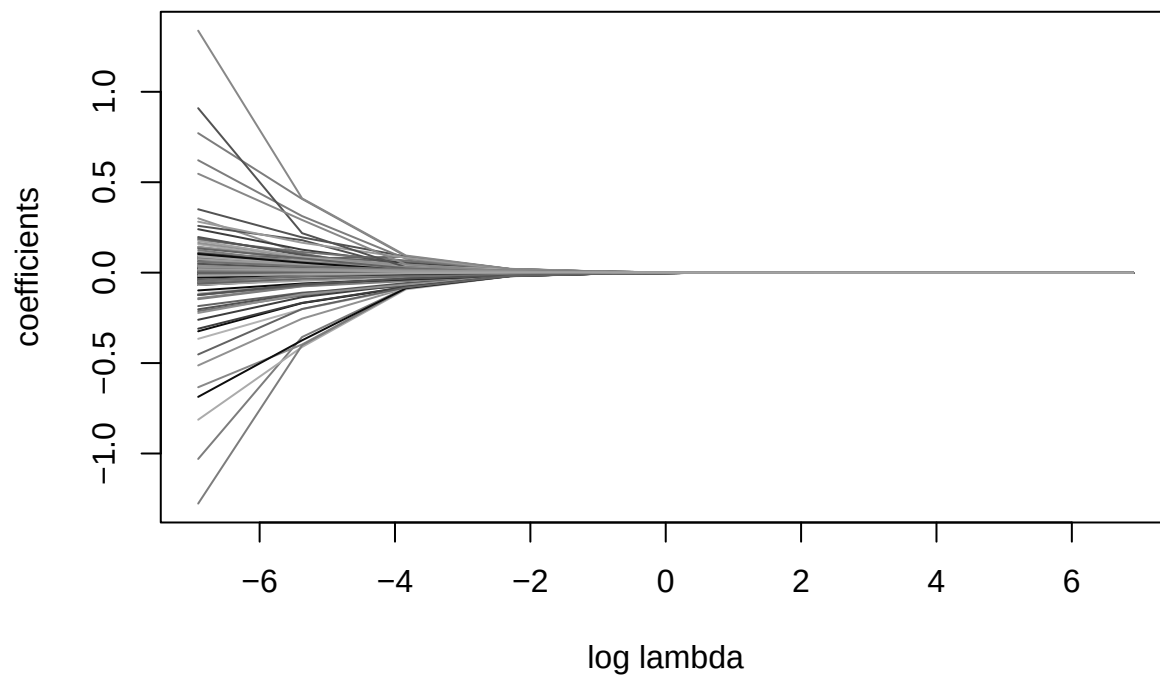


R Exercise 8: Distance Weighted Discrimination (DWD). Drug Discovery Data.

Create the first visual checks of the data or fitted objects. Use these plots to understand the structure before moving on to summaries.

Slide context: R Exercise 8: Distance Weighted Discrimination (DWD). Drug Discovery Data. We now carry out DWD classification.

```
library(kerndwd) ; library(shapes)
Drug <- list()
#standardized X
Drug$X <- xn3
Drug$y <- as.factor( group*2-1 )
lambda = 10^(seq(-3, 3, length.out=10))
kern = vanilladot()
m1 = kerndwd( Drug$X , Drug$y, kern, qval=1, lambda=lambda, eps=1e-5, maxit=1e5)
plot(m1)
```

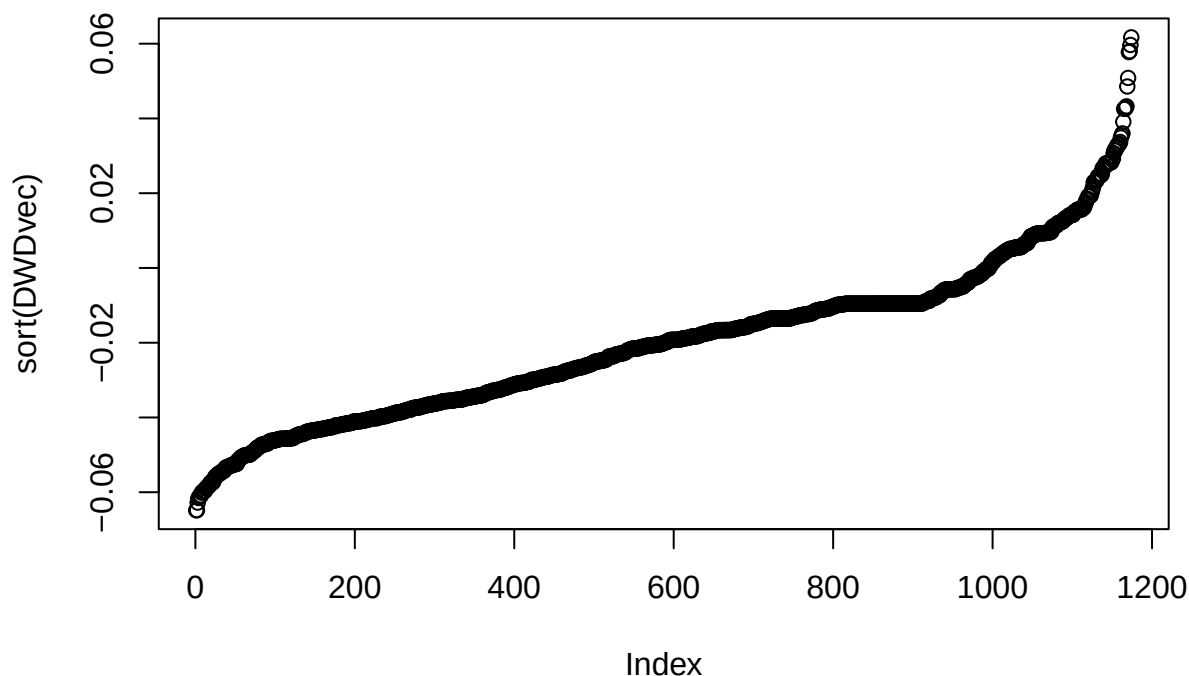
```
predict(m1, kern, Drug$X, tail(Drug$X))
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
## [257,]  1    1    1    1    1    1    1    1    1    1
## [258,]  1    1   -1   -1   -1    1    1    1    1    1
## [259,]  1    1    1    1    1    1    1    1    1    1
## [260,]  1    1    1    1    1    1    1    1    1    1
## [261,]  1    1    1    1    1    1    1    1    1    1
## [262,]  1    1    1    1    1    1    1    1    1    1
```

```
aa<-matrix(m1$alpha[2:(n+1),1],1,n)
print(lambda[1])
```

```
## [1] 0.001
```

```
DWDvec <- c(aa*%Drug$X)
DWDvec <- DWDvec/Enorm(DWDvec)
pickbig <- order( -abs(DWDvec) )[1:25]
plot(sort(DWDvec))
```



```
properties <- colnames( xn2 )
properties[pickbig]
```

```
## [1] "AAC"      "IC0"      "nHAcc"    "Jhetv"    "SEige"    "B04[C-N]"
## [7] "B03[C-N]" "B02[C-N]" "SEigm"    "SEigZ"    "B01[C-N]" "Jhetp"
## [13] "Qindex"   "B05[C-N]" "ZM1V"     "TPSA(NO)" "ZM2V"     "TPSA(Tot)"
## [19] "nN"       "SEigv"    "B06[C-N]" "SEigp"    "Ram"      "nCIC"
## [25] "F04[C-N]"
```

R Exercise 8: DiProPerm Test: Drug Discovery Data.

Use resampling or permutation ideas to assess the stability or significance of the observed result.

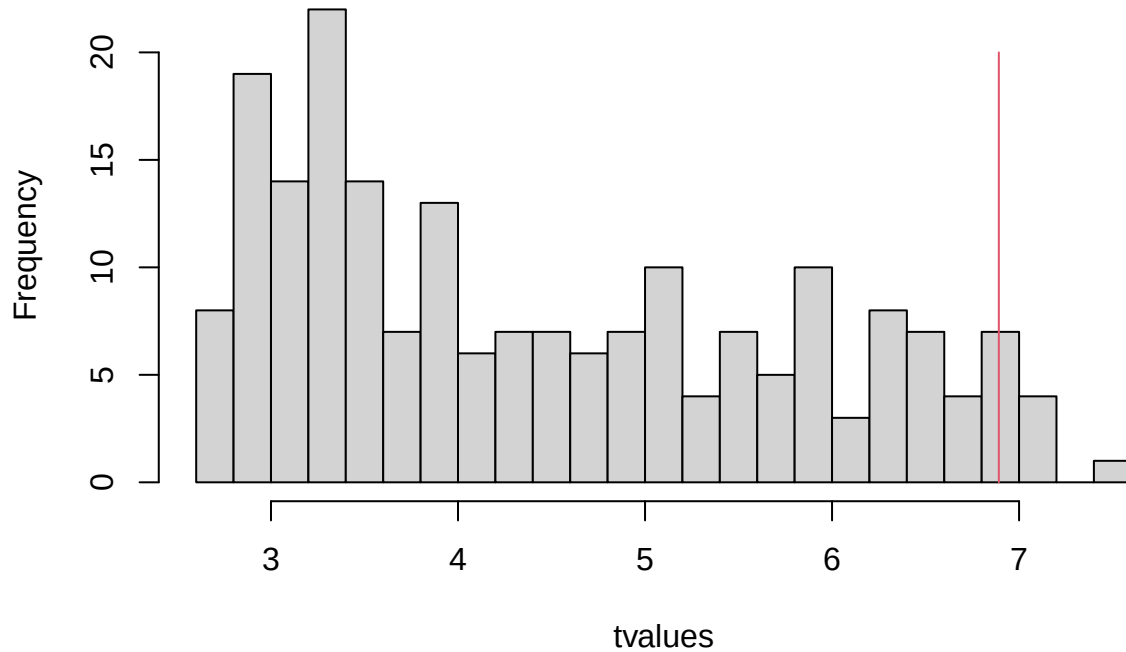
Slide context: R Exercise 8: DiProPerm Test: Drug Discovery Data. Is there a significant difference between the two groups or not? Carry out a DiProPerm test.

```
B <- 200 ; tvalues <- rep(0,times= B )
for (i in 1:B){ cat(c(i," "))
if (i==1){ Drug$X <- xn3 }
if (i > 1){ Drug$X <- xn3[sample(1:n),] }
Drug$y <- as.factor( group*2-1 )
lambda = 10^(seq(-3, 3, length.out=10))
kern = vanilladot()
m1 = kerndwd( Drug$X , Drug$y, kern, qval=1, lambda=lambda, eps=1e-5, maxit=1e5)
aa<-matrix(m1$alpha[2:(n+1),1],1,n)
DWDvec <- c(aa%*%Drug$X)
DWDvec <- DWDvec/Enorm(DWDvec)
DWDscore <- Drug$X%*%DWDvec
out <- t.test( DWDscore ~ group )
tvalues[i]<-abs ( out$statistic ) }
```

```
## 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28
```

```
obs <- tvalues[1] #observed t-statistic
hist(tvalues,nclass=30) #permutation distribution, with observed value marked in red
lines( c(obs,obs) , c(0,20), col=2)
```

Histogram of tvalues



```
print( (sum( tvalues > obs)+1)/(B) ) #pvalue
```

```
## [1] 0.045
```

R Example 9: Overlapping Classes Data

Carry out a principal component analysis and inspect the dominant modes of variation.

Slide context: R Example 9: Overlapping Classes Data Carry out PCA of the OverlappingClassesData.xlsx .
PCA

```
x <- read_xlsx("OverlappingClassesData.xlsx",col_names=FALSE)
y<-as.matrix(t(x))
group<- c(rep(1,times=50),rep(2,times=50))
n<-100
xn4 <- y
pca<-prcomp(xn4)
plot(pca$x,col=group)
```

